Draft Date: October 9, 2024

This paper has not yet been peer reviewed

Title

Investigating Causal Questions in Human Development using Marginal Structural Models: A Tutorial Introduction to the *devMSMs* Package in R

Authors

Isabella C. Stallworthy[1], Meriah L. DeJoseph[2], Emily R. Padrutt[3], Noah Greifer[4], & Daniel Berry[3]

Affiliations

1. Department of Bioengineering, University of Pennsylvania
2. Graduate School of Education, Stanford University
3. Institute of Child Development, University of Minnesota
4. Institute for Quantitative Social Science, Harvard University

Abstract

Causal inference and questions about timing are central to developmental science. Yet scientists are often limited to observational data, and conceptual, methodological, and computational barriers have prevented uptake of relevant methodological tools from other disciplines that are designed to improve causal inference. This paper provides a practical guide to implementing one powerful causal inference tool, marginal structural models (MSMs), to strengthen our ability to test causal questions about exposures/treatments that unfold over time. We draw on an illustrative example testing the potentially causal roles of dose and timing of exposure to economic strain from infancy through early childhood on children's behavior problems in early childhood. We first introduce the advantages of MSMs for investigating causal questions over time. Specifically, we review the logic of 'exposure histories' and the potential outcomes framework, using sequential randomization as a conceptual tool. We then discuss inverse-probability-of-treatment-weighting (IPTW) for addressing the threat of confounding over time. We then outline how to implement MSMs with observational data, before providing step-by-step guidance for using our novel, open-source R package, *devMSMs*. In our example, we create longitudinal IPTW balancing weights to attenuate a number of confounder-exposure relations. We then apply these weights to test an outcome model relating economic strain across infancy, toddlerhood, and early childhood to future behavior problems. Finally, we estimate and compare the predicted causal effects of different developmental histories of high economic strain that vary in dose and timing and discuss substantive implications. We close with ongoing considerations.

Keywords

Causal inference; developmental science; inverse-probability-of-treatment-weighting; marginal structural model

Developmental scientists aim to clarify the fundamental drivers of human functioning and development, to promote lifelong thriving. Central to this enterprise are questions about causality, including identifying the factors that “make a difference” (Woodward, 2005), and the timing of when these factors matter most. For instance, developmentalists are interested in how early experiences unfold over time to shape brain development, the acquisition of core skills, and psychosocial functioning in the world –often with the ultimate aim of intervening on these experiences as a means of supporting positive child development. These questions motivate a causal explanation; they require moving beyond describing associations to explaining what, when, and to what degree different factors impact development (Yarkoni, 2022). Because experiences unfold and accumulate in different ways in different individuals, research questions often concern the causal implications of different timing and dosages of experiences. Notably, addressing questions about the causal impact of factors that change over time has proven to be a vexing methodological challenge. Many experiences of theoretical and practical import (e.g., poverty) cannot be practically or ethically assigned to individuals at random at a single time point, let alone repeatedly on specific temporal schedules. Further, the utility of traditional tools for minimizing confounding bias in studies of naturally varying exposures (e.g., regression-based statistical controls) often breaks down in the context of time-varying causal processes.

Tools have been developed in other disciplines (e.g., epidemiology) to articulate and test questions concerning dosage and timing and given certain assumptions, to strengthen causal inference using observational data. However, these tools are under-utilized in psychology and neuroscience. In this paper, we introduce one such tool– Marginal Structural Models (MSMs; Robins et al., 2000). Although other excellent, accessible introductions to MSMs exist (e.g., Blackwell, 2013; Bray et al., 2006; Thoemmes & Ong, 2016; Vandecandelaere et al., 2016), we pair ours with a novel, user-friendly R package providing a prototypic MSM pipeline alongside a concrete real data example to support implementation of the method. The paper is organized as follows: In Sections 1 to 2, we briefly introduce the

potential outcomes framework for causal inference (Robins, 1986; Rubin, 1974, 2005) – specifically, the way it helps us frame questions concerning timing and dose as outcome differences due to alternative *histories* of exposure/treatments over time. We use an idealized example of repeated, sequential randomization to exposures over time to develop an intuition for these ideas. In Sections 3 to 5, we describe the challenges of applying these ideas to 'real world' observational data and how common tools like multiple regression fall short when confounding occurs over time. Drawing from Robins and colleagues (2000), we then introduce Inverse Probability of Treatment Weighting (IPTW) as an alternative tool that allows us to avoid the pitfalls of regression adjustment and, given several (sometimes non-trivial) assumptions, potentially approximate the idealized world of sequential random assignment. In turn, this allows us to revisit the causal effects of exposure histories developed previously. In Sections 6 to 7, we detail how these methods can be implemented using our novel *devMSMs* package for R. Our broad aim is to make these ideas and methods readily accessible to applied researchers entertaining causal questions concerning timing and dose. Finally, we close with discussion and concluding remarks in Sections 8 to 9.

Throughout the paper, we employ the term *exposure* (sometimes referred to as "treatment" in other literatures) broadly to encompass a variety of environmental factors, individual characteristics, or experiences that constitute putative causal events within a causal model. Exposures may be distal or proximal, reflecting a developing child's experience within different environments at many levels (Bronfenbrenner & Ceci, 1994), from the family (e.g., parenting), home (e.g., economic strain), school (e.g., teacher quality), and neighborhood (e.g., diversity) to the greater politico-cultural-economic context (e.g., inequality). Exposures could also reflect factors internal to the child, including neurodevelopmental (e.g., risk markers), physiological (e.g., stress), and psychological (e.g., anxiety) patterns, to which the child's development is exposed. An *outcome* refers to any future construct (e.g., behavior problems)

hypothesized to be causally affected by the exposure. Table 1 summarizes key terms used throughout this paper and the *devMSMs* R package.

Table 1

Definitions and Examples of Terminology

Term	Definition	Example
Exposure	Exposure or experience that constitutes the causal event of interest and is measured at least two time points, one or more of which is measured prior to the outcome.	Family economic strain Figures 2-3
Exposure Time Points	Time points when the exposure was measured, at which weights formulas will be created.	6, 15, 24, 35, 58 months
Exposure Epochs	(optional) Further delineation of exposure into meaningful units of time, each of which could encompass multiple exposure time points, that together constitute exposure main effects in the outcome model and exposure histories.	Infancy = 6 & 15 mo Toddlerhood = 24 & 35 mo Early childhood = 58 mo
Exposure Histories	Sequences of relatively high ("*h*") or low ("*l*") levels of exposure at each time point or epoch.	*"lll", "llh," "lhl," "hll," "hhl," "lhh," "hlh," "hhh"* Figure 1
Exposure Dosage	Total cumulative time points/epochs during which an individual experienced high (or low) levels of exposure, across an entire exposure history.	0, 1, 1, 1, 2, 2, 2, 3 Doses of *high* economic strain for above histories Figure 1
Backdoor Path	Any alternative, non-causal way that the exposure at a given time point associates with the outcome within a causal diagram, often via confounders.	Figure 3
Confounder	A common *cause* of an exposure and the outcome; adjusting for all confounders is required to block all backdoor paths.	Parental education Figure 3
Time-varying	A confounder possibly caused (indirectly or directly) by a prior exposure, which often reflects a construct that varies in time.	Maternal depression Figure 3
Time invariant	A confounder not caused by any exposure, which often reflects a construct that is stable over time.	Birth weight Figure 3
Collider	A variable that represents a common *effect* of two variables; adjusting for a collider problematically opens a backdoor path.	Maternal depression (*T3*) Figure 3
Balance	The degree to which confounders are independent of the exposure, possibly after adjustment by weighting.	Figure 4-5
IPTW Weights	A weight value given to an individual to orthogonalize exposure and confounders and create a weighted world.	Figure 4B
Weighted World	A synthetic population approximating sequential random assignment in which exposure is independent of measured confounders, conditional on balancing weights and assumptions (Cole & Hernán, 2008).	Figure 4B&C- 5
Outcome	Any construct measured at least once at a final outcome time point upon which the exposure is theorized to have causal effects.	Behavior problems Figures 2-3
Causal Estimand	Causal effect of interest with respect to the researcher's target population or sub-population (Greifer & Stuart, 2021).	Average treatment/exposure effect (ATE) in the population

Marginal Structural Model (MSM)	An approach for modeling and comparing the marginal distributions of potential outcomes to estimate causal relations (Robins et al., 2000).	Figure 5

Note. Cells are colored according to topic to reflect terms associated with delineating exposure (yellow), identifying confounders (orange), attenuating confounding (green), and outcome modeling (blue).

1. Causal Effects over Time

Psychological and developmental scientists are often interested in the way exposures accumulate within individuals to exert effects on more distal outcomes. For example, some developmental theories posit that there may be sensitive periods in development, in which exposures at some timepoints have comparatively more pronounced impacts on children's long-term outcomes than at others (e.g., infancy vs early childhood, e.g., Knudsen, 2004). Other models may concentrate more on the repeated, cumulative nature or 'dosage' of these exposures. For instance, models of *allostatic load* highlight the way 'repeated hits' of environmental stress lead to wear and tear on our physiological stress systems in ways that result in the gradual loss of flexible adaptation (e.g., Blair et al., 2011; McEwen & Seeman, 1999). Other frameworks may be more concerned with the simple accumulation of exposures, somewhat agnostic (or unspecific) about the sequence or developmental span in which they occur. This is implicit, for example, in *cumulative risk* models, which often tally children's putatively adverse exposures as a causative factor in children's development (e.g., Kerker et al., 2015). Of course, these examples are not mutually exclusive; there are many instances in which we expect the developmental timing, relative timing, and cumulative dosage of children's exposures to intersect in complex ways.

What they have in common, though, is that they are concerned with *exposure histories* – the way exposures unfold within individuals over time and how different exposure histories exert causal forces on development. To test causal questions about time-varying exposures, we need to identify the specific causal effects we are attempting to estimate. Thus, we begin by briefly

introducing the 'potential outcomes' framework for causal inference (Robins, 1986; Rubin, 1974, 2005) and how it applies to time-varying exposures and the comparison of exposure histories through MSMs (Robins et al., 2000). We start with a simple example of a time-invariant exposure, and then extend the logic to the ways exposures accumulate within individuals over time. Throughout, we use the effects of family economic strain on children's later behavior problems (e.g., aggression, anxiety) as our ongoing example. This is informed by theory and prior empirical work suggesting that family economic strain can undermine social affordances (e.g., parent-child interactions, predictable environments) thought to support children's abilities to regulate their emotions and behavior (Blair & Raver, 2016; Haft & Hoeft, 2017; Raver, 2012). These impacts may be particularly salient when they occur in infancy, given that the foundations of children's self-regulatory skills (e.g., sustained attention, co-regulation with caregivers) have been proposed to emerge during this early developmental span (Brandes-Aitken et al., 2019; Combs-Orme & Cain, 2006; Frick et al., 2018; Hanson et al., 2013). Thus, as we detail below, our example also implies specific hypotheses about developmental timing that will need to be addressed in our approach.

1.1 Potential Outcomes Framework

The logic of the potential outcomes framework draws from the idea that causal effects are defined by the difference between an individual's outcome given the exposure they *actually* received (i.e., factual outcome) compared to the outcome they *would have had*, had they received a different exposure (i.e., counterfactual outcome; Hernán & Robins, 2020; Imbens & Rubin, 2015; Rubin, 1974). For example, in the case of an exposure at a single time point, what would the behavior problems of a child who experienced *high* levels of economic strain look like, had the same child experienced *low* levels of economic strain instead? Of course, this is an impossible scenario, given that any single individual's outcome corresponds only to the exposure they actually received. Rubin (Rubin, 1976, 1978) described this as the 'missing data' problem of causal inference. Thus, we need to identify the causal effect by finding a stand-in for

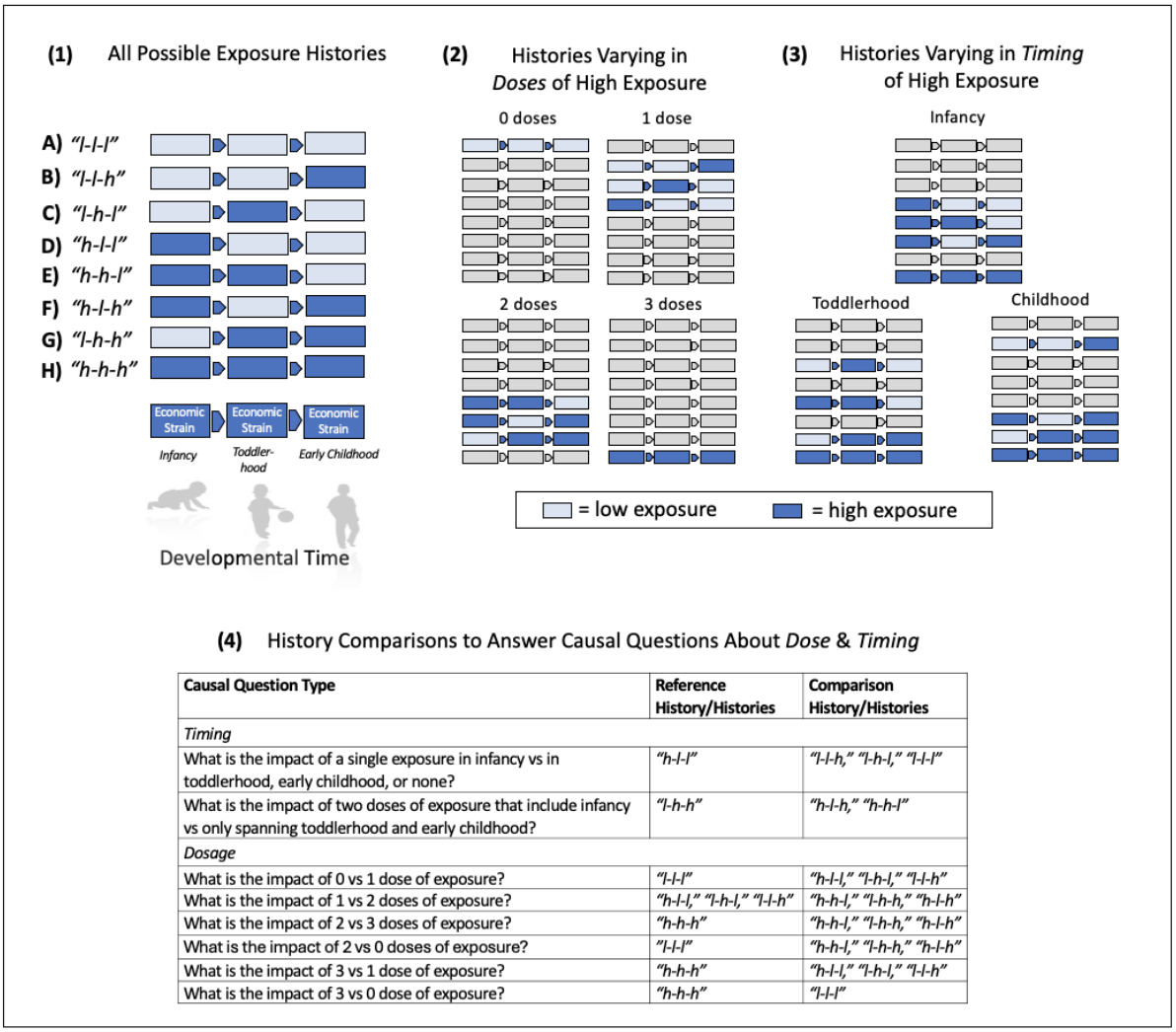
this missing potential outcome that is *as if* the same child had received a different exposure than they actually received. As we are often interested in the average causal effect for populations, as opposed to individuals, this typically means comparing the outcomes of groups of exposed and unexposed individuals who are, on average, identical –or 'exchangeable'– aside from their exposure status. Empirically, random assignment to exposure groups/levels helps to achieve exchangeability because it assures that all factors other than exposure itself are the same or *balanced* across exposure groups/levels, on average. In turn, the average observed outcome in each exposure group can serve as a reasonable average counterfactual outcome for the other, and the difference between the group averages constitutes the average causal effect of the exposure. Specifically, this would reflect a particular causal estimand, or causal effect of interest, known as the average treatment effect (ATE). The ATE refers to the difference in average outcomes if everyone were to receive one level of the exposure versus if everyone were to have received another level (see Greifer & Stuart, 2021 for detailed discussion of causal estimands).

1.2 Linking Causal Models to Exposure Histories

The same underlying logic concerning counterfactual differences applies to the causal effects of *exposure histories* –or the way exposures unfold within individuals over time (Figure 1). Yet rather than thinking about individuals being randomly assigned to a single exposure, we can engage in a thought experiment in which individuals are repeatedly, sequentially randomly assigned to an exposure group/level over time. For example, we could randomly assign individuals to high versus low levels of economic strain at infancy (*T1*), and then randomly assign them again in toddlerhood (*T2*), and again in early childhood (*T3*). Given this three time-point scenario, we can imagine several different histories of exposure to economic strain over development. A ***h***-*l*-*l* history would reflect exposure to *high* economic strain in infancy followed by *low* economic strain in both toddlerhood and early childhood.

Figure 1

Graphical Summary Depicting Exposure Histories for Answering Causal Questions about Dose & Timing



(4) History Comparisons to Answer Causal Questions About *Dose* & *Timing*

Causal Question Type	Reference History/Histories	Comparison History/Histories
Timing		
What is the impact of a single exposure in infancy vs in toddlerhood, early childhood, or none?	*"h-l-l"*	*"l-l-h," "l-h-l," "l-l-l"*
What is the impact of two doses of exposure that include infancy vs only spanning toddlerhood and early childhood?	*"l-h-h"*	*"h-l-h," "h-h-l"*
Dosage		
What is the impact of 0 vs 1 dose of exposure?	*"l-l-l"*	*"h-l-l," "l-h-l," "l-l-h"*
What is the impact of 1 vs 2 doses of exposure?	*"h-l-l," "l-h-l," "l-l-h"*	*"h-h-l," "l-h-h," "h-l-h"*
What is the impact of 2 vs 3 doses of exposure?	*"h-h-h"*	*"h-h-l," "l-h-h," "h-l-h"*
What is the impact of 2 vs 0 doses of exposure?	*"l-l-l"*	*"h-h-l," "l-h-h," "h-l-h"*
What is the impact of 3 vs 1 dose of exposure?	*"h-h-h"*	*"h-l-l," "l-h-l," "l-l-h"*
What is the impact of 3 vs 0 dose of exposure?	*"h-h-h"*	*"l-l-l"*

Note. **(1)** Graphical schematic illustrating example exposure histories for developmental exposure to economic strain, and how exposure histories can represent temporal sequences of exposure that vary by dose and timing. Depiction of all 8 possible exposure histories (A-H) across infancy, toddlerhood, and early childhood for the empirical example. As indicated by their labels, histories reflect different combinations of high (*h*; dark blue) and low (*l*; light blue) levels of exposure to economic strain across infancy, toddlerhood, and early childhood. **(2)** Graphical summaries of exposure history variation by *dosage*, highlighting exposure histories that share their total doses of exposure to high economic strain (dark blue), ranging from 0-3 doses (that vary in timing). **(3)** Graphical summaries of exposure history variation in *timing*, highlighting exposure histories that share high economic strain exposure present in infancy, toddlerhood, or early childhood (that vary in dose). Of note, the graphical summaries display simplified groupings of dose and timing, while many other permutations are possible. **(4)** Example pairing of causal questions with specific history comparisons by identifying a *reference* history (or histories) to which to compare all *comparison* histories (or history). The questions about timing illustrate tests of the

idea that infancy is a sensitive period in development during which exposure to economic strain is particularly impactful. The questions about dose illustrate several linear and nonlinear dose effects of economic strain. We recommend users select for modeling only those histories relevant to their hypotheses

Alternatively, a *l*-***h***-*l* history would reflect exposure to low economic strain in infancy and early childhood, with *high* economic strain only during toddlerhood. With three developmental time points and binary *high* and *low*, we could specify eight possible exposure histories (2^t), listed as rows A - H in Figure 1.1. If effective, repeated, sequential random assignment would render these eight possible exposure histories exchangeable. This is represented in the simple directed acyclic graph (DAG 1)[1], or causal diagram, shown in Figure 2 by the fact that each exposure has a direct causal effect on the outcome, and there is no other way for each exposure to impact the outcome other than the individual exposure itself. As such, the average observed outcome in each exposure history serves as a plausible average counterfactual outcome for the others.

In turn, just as our causal effect for the time-invariant exposure was captured by comparing two potential outcomes for a single exposure, the causal effects for timing and dose are captured by the difference between specific counterfactual *exposure histories*. In other words, the question becomes, “What is the average expected outcome for a prototypical individual who experienced some specific exposure history (e.g., *h*-*h*-*h*), and does this differ, on average, from the expected counterfactual outcome that this prototypical individual would have had, had they experienced some other specific exposure history (e.g., *l*-*l*-*l*)?” We can see how

[1] DAGs are a type of causal diagram in which researchers represent existing empirical and theoretical evidence for causal processes by drawing single-headed arrows between constructs, specifying both the presence and absence of putative causal relations, as well as assumptions that would support/contradict a causal interpretation (Greenland et al., 1999; Haber et al., 2022; Hernán et al., 2002; Pearl, 1995). Creating DAGs to represent complex real-world human processes is undoubtedly difficult; they are imperfect and subject to revision as knowledge grows, and they represent the current “best guess” about the causal process of interest based on the research team’s expert knowledge (Hernán & Robins, 2024). Online graphical user interfaces like *daggity* (Textor, 2015) can aid with the creation of DAGs.

this is captured in DAG 1 (Figure 2) by shifting our attention away from the specific exposure effect at each timepoint, to concentrate on the collective effects *across* the timepoints.

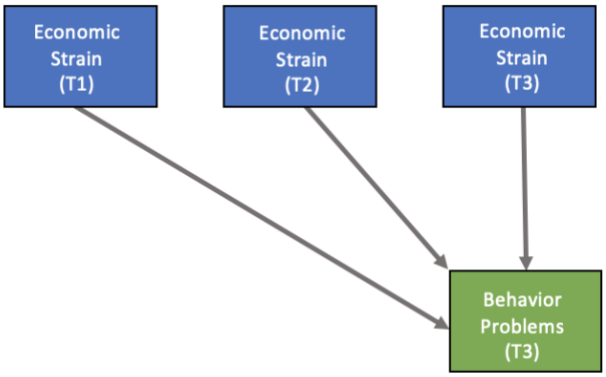
An MSM is a mathematical model relating exposure values to the expected potential outcome under that exposure history, e.g., $E[Y^{A1,A2,A3}] = f(A1, A2, A3)$ for a generic scalar function f, where $E[Y^{A1,A2,A3}]$ is the expected potential outcome for a given exposure history $(A1, A2, A3)$. Parameterizing this model and estimating its parameters allows one to estimate and contrast the expected potential outcomes under each treatment history. For example, given the exchangeability represented by DAG 1 (Figure 2) and assuming an appropriate functional form for the effects (e.g., linear and additive), one could specify the MSM using the following regression equation, where $E[BehavProb_{t3}^{ESt1,ESt2,ESt3}]$ is the expected potential outcome for a given history of exposure to economic strain (*ES*) across times *T1*, *T2*, and *T3*:

$$E\left[BehavProb_{t3}^{ESt1,ESt2,ESt3}\right] = B_0 + B_1 * ES_{t1} + B_2 * ES_{t2} + B_3 * ES_{t3}$$

Then, one would estimate parameters of the MSM by: (a) regressing the outcome on our three time points of economic strain (including, perhaps, their interactions with each other to allow non-additivity); (b) recovering the model-implied outcomes for specific exposure histories by plugging in their respective predictor values (e.g., *h-h-h* = 1-1-1 vs; *l-l-l* = 0-0-0), and (c) comparing their differences.

Figure 2

Idealized DAG Representing Causal Effects of Time-Varying Exposure on Outcome



Note. Directed acyclic graph (DAG) for conceptualizing the causal effects of exposure to economic strain at three time points (*T1*, *T2*, and *T3*) on behavioral problems, under conditions of sequential randomization.

Critically, in substantive terms, this requires the researcher to outline the specific counterfactual exposure history comparisons that are relevant to the motivating theoretical model. That is, whereas our counterfactual differences were limited in the context of a time-invariant exposure (e.g., exposed versus unexposed), time-varying exposures open up a host of possible counterfactual differences. We can see this by considering the eight possible exposure histories evident when we have three waves of exposures treated as 'high' and 'low' (as shown in Figure 1). This affords 28 potentially unique counterfactual outcome comparisons (i.e., $\frac{8!}{2!(8-2)!} = 28$). Of course, only selected comparisons will be relevant to one's specific theoretical model, highlighting why theoretical specificity is so important for working with time-varying exposures. In addition to the basic notion that causal inference using the potential outcomes framework requires specific counterfactual comparisons, the researcher also needs to establish

which of the many (here, 28) possible comparisons would support or contradict their substantive theoretical model.

2. Comparing Exposure Histories to Answer Timing & Dose Questions

Aligning one's substantive model with an appropriate ensemble of specific counterfactual comparisons is a crucial initial step for any causal question concerning timing and dose. Which comparisons are consistent, contrary, or irrelevant to the substantive hypotheses motivating the analysis? For example, conceptually, of the 28 counterfactual comparisons available to us in our economic strain example, which are consistent with the hypothesis that the long-term effects of economic strain are particularly pronounced when experienced in infancy? How can we differentiate this developmental timing hypothesis from alternative explanations concerning dose? There is no single way to approach this endeavor, and it will be informed by the precision of one's substantive model. For our working example, let us assume a modest level of specificity: (a) greater doses of high economic strain cause worse behavior problems, yet (b) the magnitude of these impacts is stronger when experienced in infancy compared to toddlerhood or early childhood. Of course, this leaves many potential questions exploratory (e.g., linearity, developmental timing effects beyond infancy, etc.). It is nonetheless a start (and a signal for areas of the substantive model that require more development).

Given our theoretical model concerning infancy as a sensitive period, we might reasonably hypothesize that the behavioral problems of children who experience a single dose of economic strain in infancy will, on average, be comparatively higher than those who experience a single dose of economic strain in toddlerhood or early childhood, as well as those who never experience economic strain. This leads to the collective hypotheses comparing the timing of a single dose of exposure (Figure 1.4):

- ***h***-/-/ (Figure 1.1 D) > /-/-***h*** (Figure 1.1 B)
- ***h***-/-/ (Figure 1.1D) > /-***h***-/ (Figure 1.1 C)

- ***h***-*l*-*l* (Figure 1.1D) > *l*-*l*-*l* (Figure 1.1 A)

A second timing question might be, does experiencing two doses of economic strain during a period that is inclusive of infancy cause a greater elevation in behavior problems compared to experiencing two doses outside of infancy? This leads to the collective hypotheses comparing the timing of two exposures (Figure 1.4):

- ***h***-***h***-*l* (Figure 1.1 E) > *l*-***h***-***h*** (Figure 1.1 G)
- ***h***-*l*-***h*** (Figure 1.1 F) > *l*-***h***-***h*** (Figure 1.1 G)

Alternatively, perhaps, developmental timing matters less than simple dosage effects. In that case, we would hypothesize similarities in behavior problems for those with exposure histories that share a dose, yet differences between exposure histories with different doses. This could lead to the following example hypotheses comparing histories of different doses of exposure to high economic strain (see Figure 1.4):

- 0 vs 1: e.g., *l*-*l*-*l* (Figure 1.1 A) vs ***h***-*l*-*l* (Figure 1.1 D)
- 1 vs 2: e.g., ***h***-*l*-*l* (Figure 1.1 D) vs ***h***-***h***-*l* (Figure 1.1 E)
- 2 vs 3: e.g., ***h***-***h***-*l* (Figure 1.1 E) vs ***h***-***h***-***h*** (Figure 1.1 H)
- 2 vs 0: e.g., ***h***-***h***-*l* (Figure 1.1 E) vs *l*-*l*-*l* (Figure 1.1 A)
- 3 vs 1: e.g., ***h***-***h***-***h*** (Figure 1.1 H) vs ***h***-*l*-*l* (Figure 1.1 D)
- 3 vs 0: e.g., ***h***-***h***-***h*** (Figure 1.1 H) vs *l*-*l*-*l* (Figure 1.1 A)

This approach of systematically aligning *a priori* hypotheses about timing and dose with specific exposure history comparisons can be challenging. Moreover, the results with real data will rarely square exactly with *all* of the theoretically motivated expectations. Thus, the researcher is left to make a case for whether the *collective* results are consistent (or not) with their substantive model. Despite these challenges, this approach affords noteworthy advantages. It supports specificity with respect to the causal inferences one is trying to make –a prerequisite of causal inference via the potential outcomes framework. It also supports specificity between one's substantive and empirical models, consistent with historical (Meehl,

1967) and contemporary (Bringmann et al., 2022; Frankenhuis et al., 2023; Muthukrishna & Henrich, 2019) calls for improving this alignment in psychology. When our substantive models are strong and precise, heightened specificity allows us to adjudicate between alternative causal models for the phenomenon of interest (e.g., sensitive periods vs dose). When the connection between our substantive models for timing and dose are still being refined, disconnects between our nascent hypotheses and empirical findings highlight *specific* areas in need of further development–a win-win scenario for cumulative science.

3. Comparing Exposure Histories to Answer Causal Questions in the Real World

For obvious ethical and practical reasons, many exposures of scientific and practical import cannot be randomly assigned to people at a single time point, let alone repeatedly over time. As such, we are left with observed data and its many challenges for making causal inferences–time-varying or otherwise. The problems arise because the assumption of exchangeability rarely holds with naturally occurring exposures. Rather, there are typically a number of confounding variables (confounders; see Table 1; Foster, 2010; Hernán et al., 2004), that can serve as alternative explanations for the effects of the exposure on outcome (e.g., DAG 2; Figure 3). Confounders can be conceptualized as 'backdoor pathways,' through which an exposure is associated with the outcome in ways other than the direct causal effects from exposure to outcome.

In the next section, we describe time-invariant and time-varying confounding and how the latter, in particular, undermines our ability to use common tools like multiple regression for testing questions concerning timing and dose. We then draw from the causal inference literature (Robins et al., 2000) to describe how estimating MSMs through Inverse Probability of Treatment Weighting (IPTWs) might –given certain assumptions– allow us to make DAG 2 (Figure 3) look more like DAG 1 (Figure 2) and, in turn, leverage the counterfactual comparisons we introduced above.

3.1 Time-Invariant Confounding

Some confounders are time invariant, as they occur only at a single time point, *prior to the exposure* and remain stable and/or[2] are not possibly affected by exposure. Sometimes these confounding relations are direct, such as in the case of parental education in our example. Extensive research demonstrates that, on average, adults with more education tend to have and retain better paying jobs that, in turn, result in less economic strain (e.g., Valletta, 2015). Thus, it is possible that parental education exerts causal effects on both the family's economic strain and children's behavior problems (Figure 3), inducing a spurious association between economic strain and behavior problems. Other times these backdoor pathways may be more indirect, like in the example of medical complications in Figure 3. Here, neither parental job insecurity nor attention dysregulation are confounders in themselves. Job insecurity causes economic strain, but not children's behavior problems. Attention dysregulation causes children's behavior problems, but not economic strain. Without a variable that simultaneously causes job insecurity *and* attention dysregulation, neither would bias the causal effect of economic strain on children's behavior problems. However, because medical complications can undermine job security as well as lead to heightened levels of attention dysregulation, it creates a backdoor pathway connecting economic strain to behavior problems.

Researchers often use regression-based approaches[3] to "partial out" the effects of confounders by including them as covariates, in attempt to isolate the relation between exposure and outcome (Foster, 2010; Thoemmes & Ong, 2016). The limitations of regression-based approaches for causal inference are well-documented (e.g., Austin, 2011; Thoemmes & Ong, 2016). For example, the results are sensitive to the specific combinations of conditioning covariates and typically assume additive, linear relations with the outcome. This can lead to

---

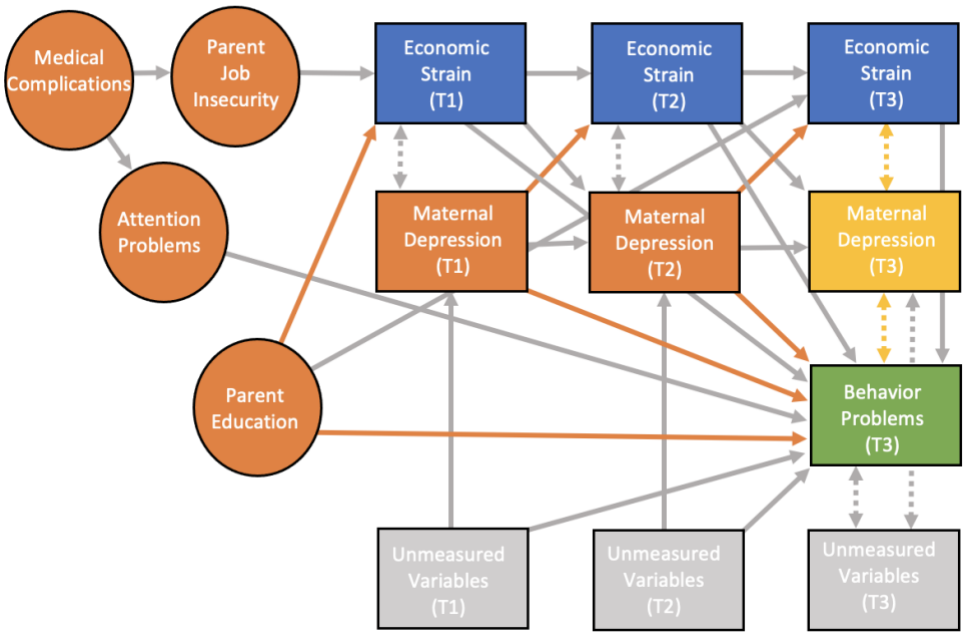
[2] Age is an example of a variable that technically varies in time but, because it cannot be affected by economic strain, would be considered a time-invariant confounder.

[3] E.g., OLS regression, generalized regression models, mixed effects models, structural equation models, etc.

overly complex models and/or strategic cherry-picking of confounders that result in substantively desirable effects (Rubin, 2001; Thoemmes & Ong, 2016). It also raises the potential for extrapolation –such that one's causal estimates are derived from adjusted data in which the distributions of the measured confounders fail to cover the entirety of the exposure distribution (i.e., based on individuals who do not actually exist in reality). Despite these limitations, given a number of assumptions, it is at least *possible* that regression-based approaches for time-invariant, pre-exposure confounder adjustment could strengthen causal interpretations. This does not hold for time-*varying* confounders, for which regression adjustment can induce more bias than it removes.

Figure 3

Abridged DAG Reflecting Real-World Causal Effects of a Time-Varying Exposure on an Outcome



Note. Time-varying exposure (e.g., economic strain) is represented by blue squares at three time points (*T1-T3*) on an outcome e.g., behavior problems) represented by a green square. In the real world, these relations are obscured by many other relations, with some examples shown here. Direct time invariant

confounders (orange circles and orange arrows, e.g., parental education) reflects a time invariant cause of both exposure and outcome, while indirect time invariant confounding (orange circle, e.g., medical complications) reflects a confounder within a causal chain extending to exposure and outcome. Time-varying confounders (orange squares and orange arrows, e.g., maternal depression) reflect those varying in time that both cause exposure and are plausibly caused by prior exposure. Direct colliders (yellow square and yellow arrows, e.g., maternal depression at *T3*) are variables that are caused by both exposure and outcome, controlling for which is problematic. As shown, time-varying confounders can also be indirect colliders when they are caused by both exposure and any unmeasured variables (gray squares, e.g., neighborhood violence) that cause the outcome, a configuration that is common constitutes a particularly pernicious challenge to regression-based adjustment strategies. Note: Contemporaneous relations (dashed lines) are shown to illustrate a theoretical causal model; the *devMSMs* package does not automatically adjust for contemporaneous confounders given that we are unable to assuredly distinguish them from mediators or colliders in the absence of a strong theoretical justification.

3.2 Time-Varying Confounding

Time-varying confounders are confounders (i.e., causes of a subsequent exposure and the outcome) that often change over time[4], even if they are not measured at every time point, and are affected by prior exposure, either directly or indirectly (Daniel et al., 2013; Robins & Hernán, 2008). For example, maternal depression could serve as a time-varying confounder for economic strain (Figure 3): depression is positively associated with financial strain (de Quidt & Haushofer, 2016; Ettman et al., 2023; Kendler et al., 1999), with some work suggesting potentially bi-directional causal effects. The strain of economic adversity can erode mental well-being, leading to depression (de Quidt & Haushofer, 2016; Hurwich-Reiss et al., 2019; Kendler et al., 1999; Price et al., 2002). In turn, depression can undermine the levels of workforce participation needed to ameliorate economic strain (de Quidt & Haushofer, 2016; Stolove et al., 2017). Evidence has also implicated maternal depression as a potential contributor to children's behavior problems (Bagner et al., 2010; Beck, 1999) (Figure 3). Thus, time-varying maternal depression could confound the effects of economic strain on behavior problems. Given this, it is

[4] For example, we consider executive functioning a time-varying confounder, given that it could reasonably be affected by prior exposure to economic strain, even if it is only measured at one time point in a dataset.

tempting to control for it in a regression model. However, this could hurt our estimate of the time-varying exposure effect more than help it (Bray et al., 2006; Foster, 2010; Thoemmes & Ong, 2016), as we described below.

First, adjusting for maternal depression amounts to controlling for a mediator in a causal path. For example, maternal depression (MD) at *T2* is a common cause of both economic strain (ES) and behavior problems (BP) at *T3* and, thus, confounds the relation between the latter. However, if we simply added *T2* maternal depression as a control covariate in a regression model, it would block the portion of the *T1* economic strain that would be transmitted indirectly through *T2* maternal depression (i.e., $ES_{t1} \rightarrow MD_{t2} \rightarrow BP_{t3}$; $ES_{t1} \rightarrow MD_{t2} \rightarrow ES_{t3} \rightarrow BP_{t3}$), biasing the effect of interest. Importantly, this same blocking problem would arise if we attempted to estimate the effects of early economic strain, while controlling for later economic strain –a common, yet flawed, practice for estimating timing effects in the developmental literature.

A second problem with adjusting for a time-varying confounder is that it may be a collider –controlling for it would induce a spurious association between the earlier exposure and the outcome (see Cole et al., 2010; Foster, 2010; Hernán et al., 2004; Robins et al., 2000). That is, in the same way that two variables can have a common cause that is a confounder, two variables can have a common *effect* that is known as a collider (Table 1, Figure 3). Conditioning on a collider (e.g., by adjusting for it in a regression model) can induce a spurious association between its causes (and/or ancestors/descendants of its causes), which may themselves be unobserved variables. For example, perhaps neighborhood violence (NV) at *T2* –an unmeasured variable in our example –causes heighted maternal depression at *T2* and worse behavior problems at *T3*. Here, one of the indirect pathways between *T1* economic strain and the outcome includes *T2* maternal depression as a common effect of both *T1* economic strain

and our unobserved variable *T2* neighborhood violence ($ES_{t1} \rightarrow MD_{t2} \leftarrow NV_{t2} \rightarrow BP_{t3}$)[5]. The problems of time-varying confounding that cannot be addressed by traditional regression adjustment are motivated the development of Robins's g-methods, which overcome these challenges (Robins, 1986). The g-methods include structural nested mean models, g-computation, and inverse probability of treatment weighting (IPTW). Below, we describe IPTW, which is the most straightforward to implement and interpret. See (Loh, 2024; Loh et al., 2024) for introductions to the former two oriented toward psychological researchers.

4. IPTWs as a Means of Approximating Sequential Randomization

Rather than attempting to directly partial out covariation shared between the exposure and confounders using regression to arrive at the 'unique' effect of the exposure, inverse-probability-of-treatment-weighting (IPTW; Cole & Hernán, 2008; Robins, 1986) attempts to remove the association between the measured confounders and the exposure at each exposure time point in an exposure history. In this way, given certain (albeit, sometimes challenging) assumptions, applying IPTWs moves us toward approximating sequential random assignment (i.e., DAG 1; Figure 2). This allows us to estimate the parameters of the MSM without inducing biases due to inappropriately adjusting for time-varying confounders, while also eliminating biases due to failing to adjust for them.

[5] Adjusting for *T2* maternal depression would open up a backdoor pathway to via *T2* neighborhood violence that would have been closed otherwise. A similar problem would arise for *T3* maternal depression ($ES_{t2} \rightarrow MD_{t3} \leftarrow NV_{t3} \rightarrow BP_{t3}$). Colliders could also be more direct. For example, even if there were no unobserved effect of *T3* neighborhood violence on *T3* maternal depression and the outcome, the ambiguity in the direction of the effect between *T3* maternal depression and T3 behavior problem raises the possibility that *T3* depression could be a common effect of *T3* behavior problems and *T2* economic strain. Controlling for *T3* depression would induce a spurious association between *T2* economic strain and the outcome. Indeed, any time there is a possibility that a potential confounder is actually caused by an exposure (i.e., is a mediator and/or collider, in actuality), adjusting for it will undermine the estimate of interest. For example, given the directional ambiguity of the relation between contemporaneously measured economic strain and maternal depression, we would need to give serious consideration to the costs and benefits of attempting to adjust the latter.

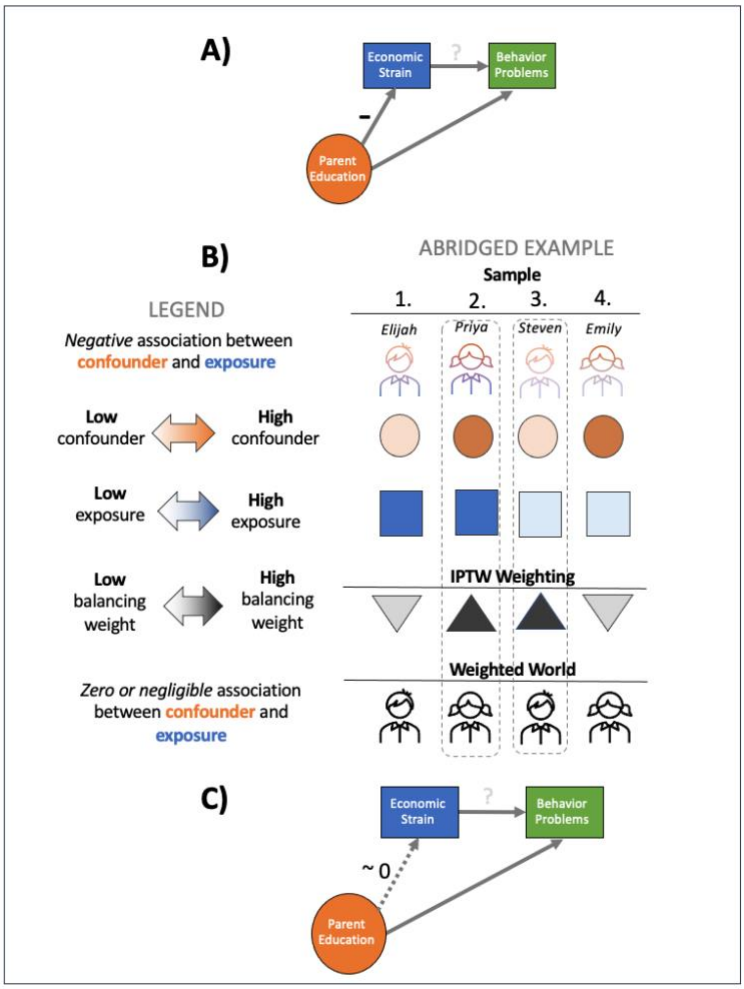
4.1 Logic of IPTW

IPTWs draw from the logic of randomization (Hernán et al., 2000; Robins, 1986; Robins et al., 2000), which, when effective, intrinsically balances all confounders across levels of the exposure, making the potential outcomes exchangeable across exposure levels. IPTWs aim to achieve the same goal –albeit, only for measured confounders. IPTW follows a similar logic to that of sampling weights. For example, any time a researcher samples from a population in a way that deviates from random sampling –for instance, by over-sampling particular subgroups– the sample is no longer representative of the population to which they aspire to make inferences. However, if we know the conditional probability of being sampled based on these subgroup characteristics, we can use these probabilities to *down*-weight the contributions of *over*-sampled individuals and *up*-weight the contribution of *under*-sampled individuals. In doing so, we create a 'weighted world' in which our weighted analytic sample is representative of the population, even when the raw data are not.

IPTWs leverage the same logic yet applied to the conditional probability of receiving the exposure, rather than the conditional probability of being sampled. Specifically, our raw data may show evidence of confounding, such that there are clear empirical associations between many potential confounders and our exposure of interest (Figure 4A). However, by estimating each individual's conditional probability of receiving the exposure given their particular collection of confounder values, we can derive weights that balance these confounders across exposure levels (Figure 4B). In doing so, we create a 'weighted world' in which we have removed the association between the confounders and the exposure (Figure 4C), even though these associations remain in the raw (unweighted) data. If we believe we have measured and balanced *all* of the necessary confounders –an admittedly challenging assumption– this weighted world approximates random assignment. Importantly, in contrast to regression adjustment for confounders, IPTWs accomplish this without consideration of the form of the relationship between the confounders and the outcome.

Figure 4

Illustrative Schematic of the Logic of Creating IPTW Weights for a Single Exposure Time Point



Note. The goal is to create a weighted world in which confounding relations are attenuated using (**A**) a simplified causal diagram reflecting *negative* associations (solid line) between a time-invariant confounder (orange, e.g., parent education) and exposure (blue, e.g., economic strain) in the observed sample. (**B**) Within the observed sample, individuals (4 hypothetical) have different levels of the confounder, with high levels of parental education represented by a darker shade of orange circle. Individuals also differ in their levels of exposure, some of whom experience high economic strain (darker blue square) and others low (lighter blue square) (binarized for this example). IPTW creates individual balancing weights that reflect the inverse of their probability of exposure given their confounder, with higher weights (darker gray triangles) reflecting individuals whose exposure level is contrary to what would be expected given their confounder level (dashed boxes). Balancing weights create a weighted world in which individuals are not more or less likely to experience the exposure based on their confounder levels, (**C**) having effectively minimized/attenuated the relation between confounder and exposure (dotted line). See Box 1 for further detail.

4.2 Estimating IPTWs to Create a 'Weighted World'

In order to estimate IPTWs (see Box 1 for a detailed example), our first aim is to quantify each individual's probability of receiving the exposure (versus not[6]), conditional on an ideally exhaustive vector of observed *pre*-exposure confounders (i.e., no mediators or colliders). This can be implemented in a number of ways, ranging from regression to complex machine-learning algorithms and optimization methods (many are available in *devMSMs* through Greifer's *WeightIt* package; Greifer, 2024b). The resulting conditional probability –also known as a 'propensity score' (Rosenbaum & Rubin, 1983) –would, thus, represent each individual's estimated probability of receiving the exposure, given her unique constellation of values on these observed confounders.

For binary and categorical exposures, IPTWs are derived by taking the inverse of the probability of receiving the exposure the individual actually received. For continuous exposures, IPTWs are derived by multiplying the inverse of the conditional probability density of the exposure by a stabilization factor equal to the unconditional probability density of the exposure, both evaluated at the exposure level actually received. When done correctly, these weights ensure that each group is representative of the full sample, as if the exposure levels had been randomized (Box 1).

[6] We use a binary exposure for didactic ease; however, a growing number of alternative strategies have been developed for ordinal, nominal, and continuous exposures and are available in the *devMSMs* R package we describe below (e.g., Hirano & Imbens, 2004; Huling et al., 2023; Imai & van Dyk, 2004).

Box 1

Understanding IPTW: A Toy Example for Binary Exposure at a Single Time Point

To make the process of creating IPTW weights concrete, we can take a binary version of our example exposure, Economic Strain (1 = exposure, 0 = none) illustrated in Figure 4. Specifically, informed by our DAG, we have identified a single confounding variable –parental education– that we think explains part of the relation between economic strain and child behavior problems. Here, we operationalize parental education as having graduated from high-school (1) or not (0) for simplicity, but all that follows would apply to a continuous education level in the same way. And, in our overly simplistic world, we feel confident that parental education (operationalized in this way) is the only meaningful confounder, and that balancing it will remove all possible backdoor pathways linking economic strain and child behavior problems.

We have a toy sample of 10 individuals (Table b1), and we see a clear difference in the educational levels between those who have experienced economic strain compared to those who have not —20% versus 60% high-school graduation rates, respectively Of the 4 individuals whose parents graduated high school, only 1 experienced economic strain (i.e., $P[ES = 1|\ Hi\ Ed] = \frac{1}{4}$ or .25). Of the 6 individuals whose parents did *not* graduate high school, 4 experienced economic strain (i.e., $P[ES = 1|\ Hi\ Ed] = \frac{4}{6}$ or .67).

In turn, we use these propensity scores to create weights by taking their inverse. For the exposed group, this is $\frac{1}{p}$, and for the non-exposed group, this is $\frac{1}{1-p}$. They differ because we are interested in the (conditional) probability of receiving the exposure *that the individual actually received*.

As shown in Table b1, in the exposed group, we see that Priya has a weight (4.00) which is considerably larger than the others in this group (1.50). This is because Priya experienced economic strain, *despite her parents' high education levels*. We see something similar for the unexposed group, such that Maya and Steven have larger weights (3.00) than the rest of the unexposed group (1.33) because they both escaped economic strain, despite their parents' low education levels. By upweighting the contributions of those who are less likely to receive the exposure they actually received based on their levels of the confounder relative to those who were already likely to receive their exposure simply by virtue of their confounder levels, we balance the means of the confounder across groups.

We can see that this is true by examining the weighted means across groups. We create weighted education means for each group by dividing the sum of the weighted education values by the sum of the IPTWs. Whereas the observed data showed notable education differences across groups, the weighted means calculated from our weighted world are identical –0.40. Thus, we have 'balanced' parental education across exposure levels. In this weighted world parental education no longer is a confounder because its link with economic strain has been broken. In turn, when we subsequently apply these weights in a weighted regression model, we know that parental education will not bias the relation between economic strain and behavior problems.

Table b1

Toy Example of a Sample of 10 Hypothetical Individuals to Illustrate IPTW

	Emily	**Jose**	**Jamal**	**Maya**	**Steven**	**Sophia**	**Elijah**	**Mark**	**Aisha**	**Priya**	
Economic Strain	0	0	0	0	0	1	1	1	1	1	
Education (Educ.)	1	1	1	0	0	0	0	0	0	1	

Pred. Prob	0.25	0.25	0.25	0.67	0.67	0.67	0.67	0.67	0.67	0.25	
IPTWs	1.33	1.33	1.33	3.00	3.00	1.50	1.50	1.50	1.50	4.00	
Weighted Educ.	1.33	1.33	1.33	0.00	0.00	0.00	0.00	0.00	0.00	4.00	
Summary Scores											
Raw Mean Education	0.60					0.20					
Weighted Mean Educ. (sum IPTWs/ sum Weighted Educ.)	0.40					0.40					

Note. They have differing levels of economic strain (exposure) and confounder (parental education). Given a hypothetical logistic model fitted estimates of B_0 = .69 and $B_{educ.}$ = -1.79, each person carries a different predicted likelihood of receiving exposure and corresponding balancing weight.

With real world data, the IPTW process alone will not necessarily assure balance on its own because the performance of IPTW depends on correct specification of the models used in the weights. Indeed, it is up to the researcher to establish that the weights have actually balanced the distributions (e.g., means, variances) of the confounders across exposure levels. As we illustrate in our applied example below, this is typically done by demonstrating the lack of association between the confounders and the exposure in the weighted sample. There are various approaches to implement IPTWs including those that rely on parametric modeling, machine learning, or optimization, and, depending on the particularities of one's specific data, some approaches may create balance more effectively than others. Trying a variety of IPTW estimation methods and choosing the one that most effectively achieves balance is often a sensible approach. As long as this procedure does not involve the outcome, there is no risk of capitalizing on chance, unlike other model building and selection procedures. In addition, although balance is our key concern for bias, it is worth noting that the variability of the IPTWs can also impact precision –a bias-precision tradeoff (Chattopadhyay et al., 2020; Chattopadhyay & Zubizarreta, 2023). Consulting the *effective sample size* (ESS) –or the size of an unweighted sample with approximately the same amount of precision as the weighted

sample– provides a sense of how the weights may impact the precision of the estimates in the final outcome model. The ESS is a function of the variability of the weights. Holding balance constant, approaches that confer a larger ESS are typically preferable.

As illustrated in Figure 5, creating IPTWs for time-varying exposures is a rather straightforward extension of the process used for time-invariant exposures –we simply repeat the process, creating IPTWs for each of the exposure periods. Each new exposure time point carries the confounder set from the prior time points and adds any new confounders as they emerge over time. Importantly, this includes the prior exposure itself (Figure 5.2). For example, consider the simple economic strain example that includes one time-invariant confounder (parental education), and one time-varying confounder (maternal depression) illustrated in Figure 3. The confounder set for *T1* economic strain would include only pre-exposure baseline parental education (which is known not to be affected by future exposures). We would not include *T1* maternal depression because it is measured contemporaneously and could potentially be caused by *T1* economic strain and, thus, possibly introduce blocking and or collider bias into our effects of interest. The confounder set for *T2* would also include baseline parental education; however, we would also add *T1* maternal depression and *T1* economic strain (i.e., prior exposure). The confounder set for *T3* economic strain would include the same set as economic strain at *T2* yet pick up *T2* maternal depression and *T2* economic strain. After establishing balance for all the confounders at each of the three time points, we would compute an $IPTW_{msm}$ for the exposure history as the product of the three IPTWs (i.e., $IPTW_{msm} = IPTW_{t1} * IPTW_{t2} * IPTW_{t3}$)[7] (Figure 5.3). Large weights are common with real-world data and can be addressed by trimming (i.e., windsorizing; see Box 2).

[7] Some recent innovations in optimization-based methods for IPTWs have begun to devise ways to estimate weights directly for the exposure histories (e.g., Imai & Ratkovic, 2014). These are exciting and promising advances, though somewhat new and under-studied.

If we (a) assume that our collective confounder set captures all of the most important backdoor pathways between the time-varying exposure and the outcome and (b) establish that the distribution of each confounder is empirically balanced across exposure level at each period, then applying these weights creates a ‘weighted world’ that approximates DAG 1 (Figure 2) – sequential random assignment. We can then apply the weights using a weighted regression model, available in most statistical platforms (Figure 5.4).

We need not worry about blocking or colliding by time-varying confounders or the later exposures because these relations have been removed by the weights. We are left to test the specific counterfactual comparisons of interest by (a) fitting a weighted regression model, and (b) using model contrasts to compare the predicted values across the specific exposure histories that were motivated *a priori* by our substantive model (e.g., sensitive period versus dosage).

Statistical inferences for these comparisons should optimally account for uncertainty in both the estimates (i.e., regression parameters, exposure history differences) as well as the IPTWs themselves. This can be accomplished through bootstrapping or through M-estimation (Boos & Stefanski, 2013; Ross et al., 2024). At present, however, M-estimation can only be readily implemented in some of the weighting procedures (see *WeightIt* documentation; Greifer, 2024b). A common alternative is to treat the IPTWs as fixed (i.e., measured without sampling error) and to use a so called ‘sandwich estimator’ (also called ‘Huber-White’, ‘robust, or ‘heteroscedastic-consistent’) to estimate the standard errors for the regression parameters and exposure history differences (the latter via the Delta Method).

Box 2

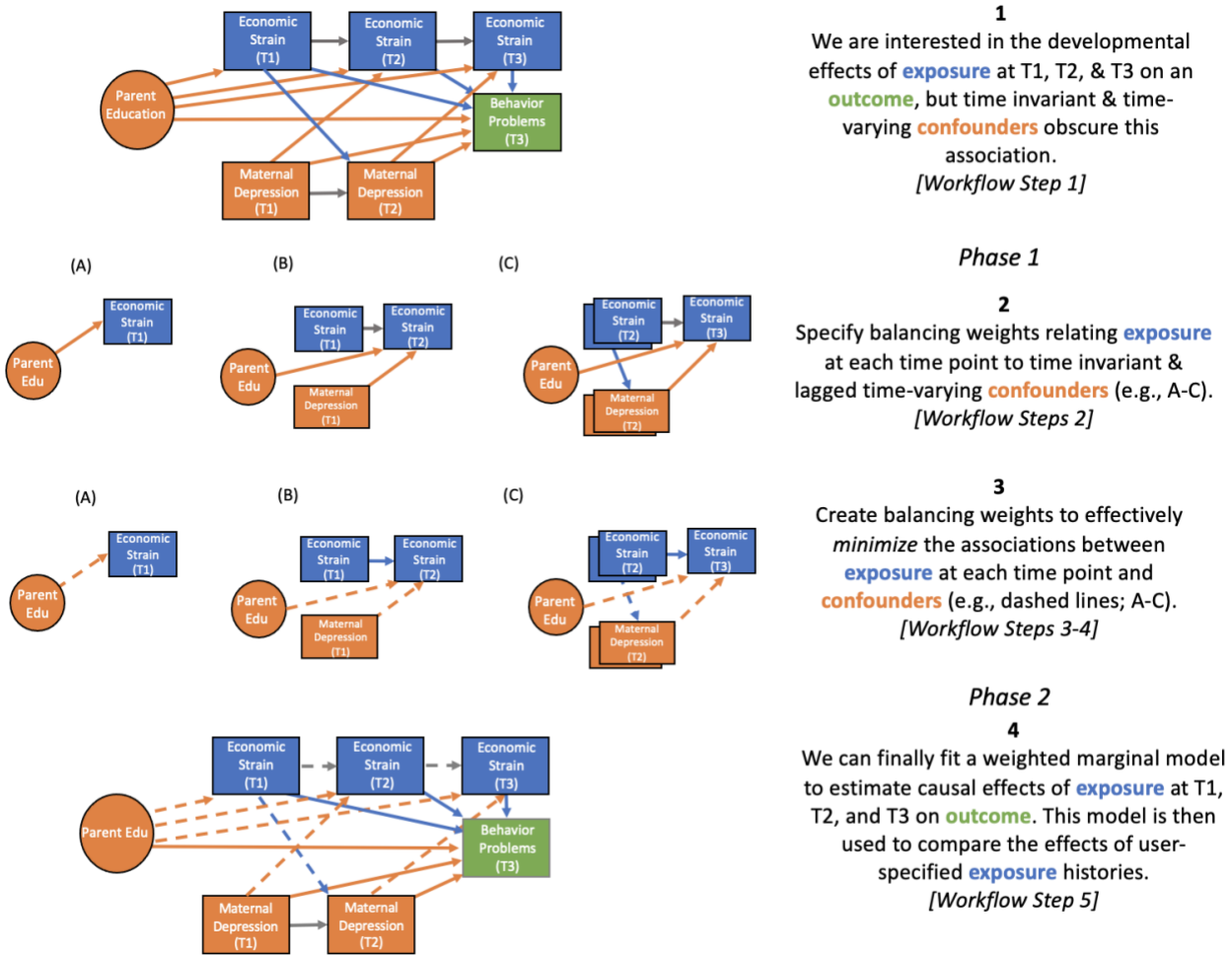
Addressing Large IPTW Weights

What if some individuals have huge weights? Depending on one's exposure and data, it is not uncommon to discover that some individuals have rather extreme weights, which can yield weights with high variability and thereby a low ESS. This occurs when an individual's collection of confounder values indicate that they have very low probability of receiving the level of exposure they actually received. Given that these individuals will have substantial impacts on both the subsequent (weighted) regression parameters, as well as the precision of these parameters, it is important to clarify whether it is sensible to treat them as outliers. For example, if someone has a near 1 conditional probability receiving the exposure, yet fails to do so, is this individual from the same population? Is it reasonable to assume that, were this a randomized experiment, this individual *could have been* randomly assigned to the control group (i.e., would comply with assignment)? If the answer is no, then it is worth considering whether including these individuals is likely to bias one's estimates and inferential statistics (see 'positivity' assumption below).

One common way to begin to address extreme weight is to stabilize them. For binary exposures, one would accomplish this by changing the numerators of inverse from 1 to the proportion of individuals in the given exposure level. For continuous exposures, IPTWs are already stabilized (Robins et al., 2000). As stabilized weights are readily implemented in *devMSMs/WeightIt*, we will forego any more detail. However, they are worth considering and typically required when working with continuous exposures (Huling et al., 2023; Naimi et al., 2014; Robins et al., 2000). In addition, there are occasions when extreme weights remain, even after stabilization. In these situations, it is common to truncate or 'winsorize' exceedingly high (99th percentile) or low (1st percentile) weights, and subsequently assess balance and ESS along with sensitivity analyses to test the implications of these analytic choices on one's substantive conclusions.

Figure 5

Summary Schematic of the Logic of MSMs via IPTW for Answering Causal Questions Involving Time-Varying Exposures



Note. We illustrate three hypothetical exposure time points (blue squares, e.g., economic strain) for one exemplar time-invariant confounder (orange circle, e.g., parent education) and one time-varying confounder (orange squares, e.g., maternal depression), in relation to an outcome (green square, e.g., behavior problems). This logic is implemented for many time-varying and time invariant confounders simultaneously. Each step is related to the steps in the proposed workflow for *devMSMs* as discussed in Sections 6 and 7, which comprise two phases of confounder adjustment followed by substantive modeling. Dashed lines reflect associations weakened by the weights.

5. Assumptions for Using IPTW to Estimate MSMs

Importantly, the validity of causal inferences arising from MSMs depends on our ability to justify a series of assumptions. The grounding logic of IPTWs for causal inference is based on the assumption that the IPTWs create a weighted world in which the exposure functions *as if it had been randomly assigned* (repeatedly so, for time-varying exposures). This is a strong assumption that rests on several other assumptions –many of which cannot be tested directly (Cole & Hernán, 2008). As mentioned above, the *ignorability* or *no unmeasured confounding* assumption is typically the most difficult to defend with confidence because it is based on the premise that the only confounders that could bias our estimates are those that we (a) measured in our study, and (b) included in our balancing model[8] (Robins et al., 2000; Rosenbaum & Rubin, 1983). Ignorability is what assures that the potential outcomes are exchangeable across exposure levels (i.e., like random assignment). This assumption gets extended in the context of time-varying exposures because we need to invoke it repeatedly –*sequential exchangeability.* Given that we never know if we have measured all the relevant confounders, this is impossible to establish (Hernán & Robins, 2024). However, for the potential confounders we have measured, DAGs are useful for establishing which set of measured confounders are likely sufficient for removing backdoor pathways between our exposures and outcomes. Ultimately, the validity of the causal interpretation hinges on one's ability to reasonably defend sequential exchangeability. Importantly, even in cases where we are unable to perfectly achieve sequential exchangeability, IPTW *still allows us to control for time-varying confounders without introducing bias* in ways that simply cannot be done using regular regression.

In addition, even if we have measured and balanced a sufficient set of confounders, another assumption is that our balancing model reasonably captures the data generating

[8] Although this is sometimes framed as including *all* possible confounders, it's more accurate to think about it as a *sufficient set* of confounders to block all backdoor connections between the exposure(s) and the outcome (i.e., because some confounders block the influence of other confounders, not all of them need to be balanced).

process that connects the confounders to the exposures (e.g., functional form of the relations, additivity, linearity, population homogeneity, etc.). Applying a simple ‘off-the-shelf’ regression model to estimate the main effects of a vector of confounders may or may not approximate the data generating process, and the validity of the IPTWs depends on the extent to which it does. Many new nonparametric, data-driven approaches impose fewer assumptions on the models used to estimate IPTWs and using them may strengthen the assumption that the relations between the confounders and exposure have been adequately captured. Assessing and demonstrating balance is what allows one to determine whether the estimated models are “close enough” to the ideal models.

The *positivity* or *overlap* assumption holds that each individual has a non-zero probability of receiving each level of the exposure, given their confounders and prior exposure history (see Zivich et al., 2022). This ensures that the exposure effects do not extrapolate. It also has practical implications, in that violations would typically lead to extremely high weights (see Box 2), which can reduce precision. Unlike some of the other assumptions, one can get a rough sense of positivity by considering the overlap of propensity scores across exposure groups (they should overlap completely for an ATE estimand), checking mean and distributional balance for the confounders, and conducting other tests that hint at potential non-overlap (e.g., Kolmogorov-Smirnov tests). With continuous treatments this can be more involved; for example, plotting the overlap of the generalized propensity scores across the entirety of the respective exposure and confounder distributions.

Finally, the *consistency* assumption concerns the idea that the observed outcome accurately reflects the potential outcome for the exposure received. Conceptually, this means that the observed outcome is what was caused by the exposure condition. For example, there was no subgroup of individuals who actually experienced high levels of economic strain yet have invalid observed scores because they understood the construct by a different name or maintained a different scaling. It means that the potential outcomes for a given child do not

depend on the exposure status of other units in the sample. For example, one child's exposure to low economic strain should not influence the behavior problems of another child experiencing high economic strain.

6. Putting it all Together –Implementing MSMs via IPTW

In summary, MSMs are a useful but underutilized tool for answering causal questions about time-varying exposures in human development. Using IPTW to estimate MSMs (under the aforementioned assumptions), researchers can gain more confidence that they are getting closer to approximating sequential randomization by accounting for a wide variety of time-invariant and time-varying confounding variables, without the limitations of regression. Collectively, this technique offers a parsimonious way of evaluating the effects of theoretically relevant doses and timing of exposures on outcomes. Below, we present a concise conceptual overview of implementing IPTW, followed by a practical step-by-step guide using our newly developed R package, *devMSMs* (Table 3). For clarity, we break down the process into a preliminary phase and two key implementation phases.

Aspects of the preliminary Phase 0 are largely reflective and practical processes critical to ensuring the subsequent implementation phases of estimating MSMs are executed accurately. As illustrated in the preceding sections, researchers must use their substantive causal model to select exposure histories and comparisons that are appropriate for addressing their hypotheses. This involves creating a DAG for the theorized causal process and mapping this model to available empirical data. Using their DAG, researchers should identify all the potential measured time-invariant and time-varying confounders (and differentiate them from colliders).

As summarized in Figure 5, Phase 1 entails adjusting for the confounders identified from the preliminary phase. The goal of this first phase is to estimate IPTW weights for each participant to balance all potential time-invariant *and* time-varying confounders *for exposure at*

each time point. The success of the balancing is evaluated by examining the remaining, post-balance associations between confounders and exposure. There is no single agreed-upon threshold for 'acceptable' imbalance. Generally, it is common to see standardized exposure-control mean differences of less than .10 or exposure-covariate correlations of less than .10 considered reasonably balanced (Stuart, 2008).

Phase 2 (Figure 5.4) entails using the weights generated in Phase 1 in the context of a weighted regression model to test substantive questions regarding the causal effects of the time-varying exposure on the later outcome. From our final outcome model, we can use the individual regression parameters to estimate the potential outcomes for each of our specified exposure histories. We can then compare outcomes between our specified exposure histories to answer our research questions about the causal effects of dose and timing.

7. Using the *devMSMs* R Package

The core motivation behind our package is to make MSMs via IPTWs approachable for applied developmental researchers. Indeed, there are already numerous excellent open source software packages for creating and assessing balancing weights and fitting weighted models (e.g., Fong et al., 2018, 2019; Greifer, 2024b) However, the iterative computation required to implement these packages with longitudinal data is non-trivial –particularly, when faced with the complexities of 'real' data (e.g., imputation to address missing data, varying exposure types, many potential confounders, etc.). Our open-source R package aims to reduce these barriers by optimizing the workflow and use of these packages –in particular, Greifer's *WeightIt* (Greifer, 2024b) and *cobalt* (Greifer, 2024a) and Arel-Bundock's *marginaleffects* (Arel-Bundock et al., 2024) packages –to support greater automatization of the process. To implement the (1) confounder adjustment and (2) substantive modeling phases of MSM-building, we provide an overview of the five practical steps (Table 3) to implement our package's seven functions (Table

2). The functions, optional helper functions, and R notebooks are elaborated upon on the package website (https://istallworthy.github.io/devMSMs/index.html).

This guide is intended to serve as a high-level overview of package implementation. Below, we describe the core workflow (Table 3) involved in using the *devMSMs* package to fit MSMs to longitudinal data, drawing from our empirical example investigating the causal effects of economic strain on behavior problems. Each section below contains a brief overview of the workflow steps and their associated *devMSMs* functions followed by their application to our empirical example. As illustrated, some steps in this workflow are automated by the functions, while others require manual user input and decision-making. We advise users to carefully inspect all automated output from the functions to ensure accuracy.

The Supplement provides more detail and code for using *devMSMs* with the empirical example. For a detailed, step-by-step guide to implementing the package as a new user, please see the vignettes and R markdowns on the package website. The website contains more information about all functions and a more detailed description of each step with full code examples that include all required and optional user specifications.

Table 2

Summary of the 7 Core Functions in the devMSMs Package

Function	Purpose	Required Input	Output
`initMSM()`	Creates an MSM object with core variables for use in all other functions.	exposure time invariant confounders time-varying confounders	msmObject
`createFormulas()`	Creates weights formulas at each exposure time point relating exposure to confounders.	msmObject type ('full', 'short', 'update')	List of weights formulas for each exposure time point
`assessBalance()`	Assesses confounder balance at each exposure time point according to best practices (Jackson, 2016).	data msmObject	List of balance statistics Tables & love plots displaying balance statistics at each exposure time point
`createWeights()`	Creates balancing weights for each person (*WeightIt's* `weightitMSM()`).	data msmObject	Weights object Histogram & summary of balancing weights
`trimWeights()`	Trims balancing weights to account for heavy tails (*WeightIt's* `trim()`).	weights	Trimmed weights object Histogram & summary of trimmed weights
`fitModel()`	Fits user-selected, weighted marginal model relating exposure to outcome.	data msmObject outcome weights model ("m0-3")	Fitted model object Omnibus test & table of model evidence
`compareHistories()`	Estimates, compares, & visualizes model-predicted outcomes as a function of exposure history.	data fitted model	Tables of estimated values and comparisons Boxplot

Note. The *devMSMs* functions can be found at the following Github: https://github.com/istallworthy/devMSMs.

Table 3

Summary of Recommended Workflow for Using the devMSMs Package to fit MSMs with Longitudinal, Observational Data

Step	#	Description	Associated Functions
Phase 0: Preliminary Conceptual and Data Preparation			
Link substantive model to MSM comparisons	P1	*Formulate Hypotheses* Identify causal of interest and accompanying exposure histories needed to test a set of hypotheses about dosage and timing.	*N/A*
Identify confounders in causal model	P2	*Create DAG* Using the DAG, identify all time varying and time-invariant confounders based on available empirical data.	*N/A*
Specify confounders, exposure, outcome, histories, confounder importance	P3	*Specify Core Inputs* Specify core package inputs based on the empirical dataset.	*N/A*
	P4	*Data Preparation & Inspection* Prepare data (format, transform, impute) to establish complete data in wide format before creating an MSM object & inspecting exposure histories & data.	`formatLongData()`[a] `formatWideData()`[a] `imputeData()`[a] `initMSM()` `inspectData()`[a]
Phase 1: Confounder Adjustment			
Estimate weights *Assess balance*	1	*Create Full weights formulas & Conduct Pre-Balance Checking* Create full formulas (i.e., with all time-varying confounders), at each exposure time point, & conduct exploratory pre-balance checking.	`createFormulas()` `assessBalance()`

	2	*Create Simplified weights formulas & Determine Optimal Weighting Method*
		Create simplified formulas (i.e., with only lag of t - 1 time-varying confounders), at each exposure time point, to create balancing weights and determine the optimal weighting method.
			`createFormulas()`
			`createWeights()`
			`assessBalance()`

	3	*Create Updated Formulas & Re-Specify Weights Using Optimal Weighting Method*
		Assess balance of optimal weights, update (add any imbalanced confounders at lag of $t > 1$) simplified formulas, and create, trim, & plot final balancing weights.
			`assessBalance()`
			`createFormulas()`
			`createWeights()`
			`trimWeights()`

	4	*Conduct Final Balance Assessment* Assess balance of final balancing weights.	`assessBalance()`
Phase 2: Assess Substantive Associations between Exposure and Outcome			

Fit weighted outcome model *Compute & compare histories*	5	*Fit Weighted Outcome Model & Summarize & Visualize Results* Select and fit a flexible weighted outcome model before estimating, comparing, & visualizing model-predicted outcome as a function of exposure histories.	`fitModel()` `compareHistories()`

Note. [a] These helper functions are not part of the *devMSMs* package, but are available at: https://github.com/istallworthy/devMSMsHelpers

Phase 0: Preliminary Conceptual and Data Preparation

As illustrated in prior sections, the first preliminary steps involve formulating hypotheses (*P1*) and creating a DAG (*P2*). Subsequently, the user identifies an empirical dataset, then specifies core package inputs (e.g., exposure, outcome, time invariant confounders, time-varying confounders), any exposure epochs, a cutoff for demarcating high and low levels of continuous exposures, and exposure histories relevant to hypotheses (*P3*), before preparing and inspecting their data (*P4*).

To examine the causal effects of economic strain across development on behavior problems, we draw from the Family Life Project (FLP) dataset. FLP is a longitudinal study following 1,292 families representative of two geographic areas (three counties in North Carolina and three counties in Pennsylvania) with high rural child poverty (Burchinal et al., 2008; Vernon-Feagans et al., 2013). This empirical example uses data collected when the children were 6, 15, 24, 35, and 58 months of age. We included a wide variety of variables in our dataset, spanning constructs at multiple levels (i.e., behavioral, physiological, demographic) from multiple sources (i.e., parent, observer, child), to encompass all available constructs assessed at these time points that may confound the relation between economic strain and behavior problems (see Supplement). To measure economic strain, we draw upon the six-item caregiver-reported Economic Strain Questionnaire (Conger et al., 1994). Specifically, we used previously generated factor scores adjusted for measurement non-invariance (see Supplement

and DeJoseph et al., 2021). Behavior problems were indexed via a mean score across items assessing emotional symptoms, conduct problems, hyperreactivity, and peer problems from the parent-reported Strengths and Difficulties Questionnaire (SDQ; see Supplement and Goodman, 2001). We note a limitation in our study design that is common with real-world data: the time point at which the behavior problems outcome is measured is the same time at which *T3* of our time-varying confounders and exposure are measured. Further, given that our empirical data had missingness reasonably assumed to be missing at random (MAR), we created 5 imputed datasets (see Supplement and the website's *Data Requirements & Preparation* vignette for more details about formatting).

P3. Specify Core Inputs

First, we identify core inputs that will be used by the *devMSMs* functions. We specify the following core inputs, with their corresponding field names that will eventually be supplied to `initMSM()` shown italicized in parentheses (see the website *Workflow* vignettes and ExampleWorkflow.rmd for more details about other optional inputs):

Required: Exposure, Outcome, & Confounders*.* Users are required to specify measured exposure and confounders. In our empirical example, we specify "ESETA1.6," "ESETA1.15," "ESETA1.24," "ESETA1.35," "ESETA1.58," (*exposure)* reflecting economic strain at 6, 15, 24, 35, 58 months. We included 32 confounders, 18 time-invariant (*ti_conf*) and 14 time-varying (*tv_conf*) (see Supplement Table s1). We identify "StrDif_Tot.58" as our outcome reflecting children's behavioral problems at 58 months.

Optional: Exposure Epochs. The user has the option to specify exposure epochs, or groupings of exposure time points, to be used in creating exposure histories and modeling the relation between exposure and outcome (Step 5). For example, in our empirical example, given that economic strain was measured at multiple timepoints during infancy, toddlerhood, and early

childhood, we specify exposure epochs[9] (*epoch*). Specifically, we specify 'infancy' corresponding to economic strain exposure at 6 and 15 months, 'toddlerhood' corresponding to economic strain exposure at 24 and 35 months, and 'childhood' corresponding to economic strain exposure at 58 months. We chose to use three epochs rather than the five raw observation time points for both substantive and practical reasons. Substantively, current theory lacks the precision to inform clear developmental hypotheses at the more granular temporal scale. At a more practical level, this weakness comes into sharp relief when one considers the exponential relation between the number of exposures and number of possible exposure histories. Thus, decisions concerning whether, when, and how to create epochs from time-varying exposures should be informed by the constraints of both theory and available data. If no epochs are specified, outcome modeling and exposure histories will be conducted using all of the time points at which the exposure is measured.

Required for Continuous Exposures: Identify High/Low Cutoff Values. When the exposure is binary, we can easily identify the presence versus absence of exposure. However, if the user has a continuously distributed exposure –as we do in our current example –one needs to specify what constitutes high versus low at each timepoint. We do this by specifying percentile values that represent high and low cutoff values (*hi_lo_cut*). These percentiles will be used in the recommended preliminary inspection step and in the modeling and comparing of exposure histories in relation to the outcome (Step 5), and subjected to sensitivity analyses. We advise balancing substance and the practical constraints of the data when making these decisions. We suggest selecting percentile values for high and low levels of a continuous

[9] For example, using a binary 'high vs low' cut off for the exposure at each time point, 3 exposure waves manifests in 8 exposure histories and 28 possible pairwise contrasts; 5 exposure waves manifests in 32 exposure histories and 496 possible pairwise contrasts. Given that MSMs concern the comparison of specific exposure histories, we would require considerable precision in our hypotheses to know which of the possible 496 contrasts are most consistent with (and contradictory to) our theoretical model.

exposure and visually inspecting the distribution of your sample across the exposure histories to ensure reasonable distribution.[10]

In our example, we have three epochs of our economic strain exposure and first selected the 75th and 25th percentiles as representing high- and low-level economic strain at each time point. However, we subsequently adjusted this to the 60th and 30th percentile (Table 4) after observing the sparsity in the sample induced by the 75th and 25th percentiles (Supplement Table s6).

Recommended: Specify Hypotheses-Relevant Exposure Histories*.* Next, we recommend the user specify exposure histories that correspond to their substantive hypotheses. If the user has specified exposure epochs, the exposure histories will consist of high and low levels of exposure across these epochs (e.g., infancy, toddlerhood, childhood). If the user has not specified exposure epochs, the exposure histories will consist of high and low levels of exposure across the exposure time points (e.g., 6, 15, 24, 35, 58 months). We recommend users specify a reference history or histories (*reference*) to which all comparison histories will be compared. Accordingly, we also recommend users specify a comparison history or histories (*comparison*) that will be compared to the reference history. These will be used in the subsequent recommended preliminary step as well as in the modeling and comparing of exposure histories in relation to the outcome (Step 5). For our example, we specify the histories denoted in Figure 1.4 that reflect substantive questions about the role of sensitive periods versus dosage of exposure to economic strain on children's behavior problems.

[10] Ideally, this should be informed by theory; however, the user must also choose values that facilitate exposure histories with reasonable representation in the data. For example, if one selected the 90th and 10th percentiles as 'high' and 'low', respectively, and there was some rank-order stability in the exposures over time, then the representation of those who are 'high' at one epoch yet 'low' at another within the dataset will be quite sparse (e.g., *h-l-h*; Figure s3). Thus, in Phase 2, when these exposure histories are used to estimate the causal effects, the estimates will be quite imprecise (i.e., large standard errors). The goal is to choose 'high' and 'low' values of the exposure that are both substantively meaningful and allow for reasonable representation in the multivariate point cloud of the time-varying exposure, in order to maximize precision for the estimates of interest.

P4. Data Preparation and Inspection

Required: Data Preparation. *devMSMs* requires complete data in wide format, as a single data frame of complete data or imputed datasets[11] either in *mids* format (output from the *mice* package; Buuren & Groothuis-Oudshoorn, 2011) or as a list of imputed datasets. We provide a series of helper functions (*devMSMsHelpers* helper package) to assist the user in preparing their data for use in *devMSMs*, including formatting (Supplement Figure s1) and imputing data to account for missingness (detailed further in the website *Data Requirements & Preparation* vignette).

Required: Create MSM Object. Having formatted and imputed our data, we supply our *mids* data and the core inputs specified above in P3 to the `initMSM()` function to create an MSM object. The MSM object will be supplied to all subsequent functions.

Recommended: Inspect Exposure Histories & Data. For the final part of this preliminary step, we strongly recommend users leverage the `inspectData()` function from the *devMSMsHelper* package to inspect their data and exposure histories. This provides a tool to help the user summarize their data (Supplement Tables s2 - s5; Figure s2) and acquire a general sense of the representation of individuals in the multivariate point cloud of the exposures over time (i.e., each exposure history; Supplement Figure s3). This is particularly useful when choosing ‘high’ and ‘low’ thresholds for continuous exposures because it helps assure that one’s chosen thresholds do not represent an overly sparse region of the multivariate exposure distribution (i.e., limited individuals representing particular exposure histories).

[11] As is recommended (Leyrat et al., 2021), all steps of the workflow are conducted separately for each imputed data before pooling results using Rubin’s rules when conducting history comparisons.

Table 4

Example Summary of Sample Distribution Across User-Specified Exposure Histories

HISTORY	n
"h-h-h"	244
"h-h-l"	12
"h-l-h"	23
"h-l-l"	25
"l-h-h"	28
"l-h-l"	22
"l-l-h"	12
"l-l-l"	146
TOTAL N	512
TOTAL %	40%

Note. Exposure histories are based on any optional user-specified exposure epochs (e.g., infancy, toddlerhood, childhood) and, for continuous exposures, user-specified cutoffs for high and low levels of exposure (e.g., 60th and 30th percentiles). Output for imputation 1. Adapted output from the `inspectData()` helper function and `compareHistories()` *devMSMs* function. This table should only be used for heuristic purposes to examine which histories will be estimated with the most precision; all units are used in all subsequent analyses, and their actual contributions depend on their estimated weights.

Phase 1: Confounder Adjustment

Having specified our causal model, selected hypothesis-driven exposure histories, prepared core inputs, and inspected our histories, we can begin the main MSM workflow. The goal of the first phase of the MSM computation is to create IPTW balancing weights that successfully minimize all associations between confounders and exposure at all time points to near zero, thereby preventing confounding of the relation between exposure and outcome. This first phase creates several kinds of balancing equations, also known as formulas, for specifying IPTW weights and assessing their performance, as summarized in Table 5. We can use the *devMSMs* functions to help automate this process.

Step 1. Create Full weights formulas & Conduct Pre-Balance Checking

The goal of this step is to use the `createFormulas()` function to automatically (*1a*) create full, comprehensive weights formulas (see Table 5) relating exposure to all measured confounders, at each exposure time point. Weights formulas will be automatically created at each of the exposure time points in the *exposure* field of the MSM object (Step P3). All confounders supplied in the MSM object will be considered. These formulas are then used to (*1b*) explore pre-balance, or existing imbalance in the data prior to weighting.

1a. Create Full Weights Formulas Relating Exposure to all Confounders at Each Time Point. We first create comprehensive, full weights formulas using the `createFormulas()` function setting *type* = "full." For each exposure time point, the function automatically outputs a formula modeling the exposure as a function of all user-specified time-invariant confounders (e.g., state) and time-varying confounders (e.g., income) at all lags prior to that exposure time point (Table 5). Time-varying confounders consist of those supplied by the user in the MSM object as well as exposure at prior time points. (See Supplement Table s7 for full weights formulas from the empirical example.) This function outputs full formulas at each exposure time point for user inspection.

1b. Conduct Exploratory Pre-Balance Assessment. Using these full formulas, we next examine the nature and magnitude of initial imbalance in the dataset, prior to weighting, using the `assessBalance()` function. This function assesses imbalance in the sample and outputs summary balance statistics and plots (Figure 6), at each time point and averaged across time points, in accordance with best practices for assessing balance with time-varying exposures (see Supplement and Jackson, 2016). For continuous exposures such as economic strain, balance is assessed by the magnitude of the correlation between the exposure and each confounder. For binary exposures, balance is assessed by the standardized mean difference for

each confounder. From the `assessBalance()` function, the user can inspect full and summarized tables reflecting confounder balance, as well as summary love plots (e.g., Figure 6; Supplement Figure s4).

Users have the option to provide a balancing threshold to the `assessBalance()` function via *balance_thresh*, a number which reflects the absolute value correlation (or standardized mean difference) between exposure and confounder below which the relation is considered negligible (the default is 0.1). This will be used by the function to determine which confounders are balanced in relation to exposure, at each time point. Further, and as recommended, users have the option to specify two balancing thresholds: a more stringent one to which the most critical confounders will be compared and a less stringent one to which less critical confounders will be compared. Here, the user also specifies to the function via *imp_conf* a list of most critical confounders (time invariant or time-varying), determined by theory and existing literature. We strongly recommend users stay consistent with their balance threshold specification throughout the use of the package.

In our empirical example, we have identified the most critical confounders as time-varying income and parent education, which we specify are only considered balanced with respect to exposure if they exhibit a correlation coefficient with an absolute value below 0.05. All other confounders will be considered balanced with an absolute correlation coefficient value below 0.1 (Austin, 2019; Stuart, 2008). We found that 52 (20%) of confounders across time points were imbalanced with respect to exposure, with a median absolute value correlation of 0.12 (max = 0.31), for imputation 1 (see Supplement Table s8 for all pre-balance statistics averaged across imputed datasets).

Table 5

Terminology Used for Automatically Generating Weights Formulas to Create IPTWs

Weights Formula Mathematical equation created at each exposure time point, regressing exposure on confounders, for the purpose of creating IPTW balancing weights $ESETA1_t = time\ invariant\ confounders + time\ varying\ confounders\ (including\ lagged\ exposures + lagged\ outcomes)$	
Term	**Definition**
Full weights formula	Weights formula containing all time invariant and *all* lagged time-varying confounders and lagged exposure values. Used for pre-balance (Step 4b). *ESETA1.24 ~ B18Raw.15 + B18Raw.6 + BioDadInHH2 + CORTB.15 + CORTB.6 + DrnkFreq + ESETA1.15 + ESETA1.6 + gov_assist + HOMEETA1.15 + HOMEETA1.6 + HomeOwnd …*
Simplified weights formula	Weights formula containing all time invariant and *only t-1* lagged time-varying confounders and lagged exposures. Used for creating preliminary balancing weights to determine optimal weighting method (Step 5b). *ESETA1.24 ~ B18Raw.15 + BioDadInHH2 + CORTB.15 + DrnkFreq + ESETA1.15 + gov_assist + HOMEETA1.15 + HomeOwnd…*
Updated weights formula	Weights formula containing all time invariant confounders, all time-varying confounders at *t-1*, as well as *any imbalanced time-varying confounders and lagged exposure levels at t-1+*. Used to create final balancing weights (Step 6). *ESETA1.24 ~ B18Raw.15 + BioDadInHH2 + CORTB.15 + DrnkFreq + ESETA1.6 + ESETA1.15 + gov_assist + HOMEETA1.15 + HomeOwnd…*

Note. Wights formulas are equations regressing exposure on confounders using the `createFormulas()` function (Steps 1a, 2a, 3b). The different kinds of formulas vary with respect to their inclusion of time-varying confounders. Each term (i.e., "full," "short," "update") corresponds to a different *type* specified to the function. Abridged example formulas for exposure to economic strain (i.e., ESETA1) are shown. Non-abridged examples and a data dictionary are available in the Supplement Table s1.

Step 2. Create Simplified Weights Formulas & Determine Optimal Weighting Method

The goal of this second step is to (*2a*) create simplified, more parsimonious weights formulas to (*2b*) create IPTW weights using all available weighting methods, before (*2c*)

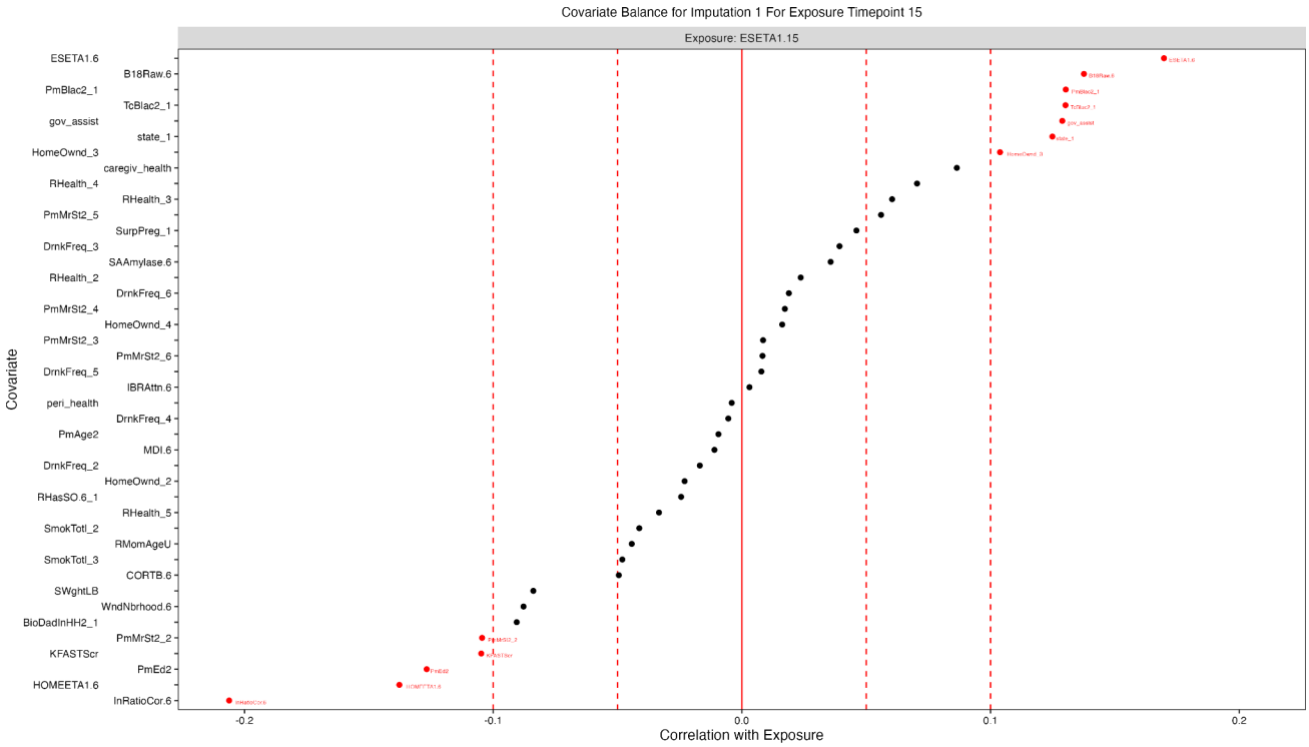
assessing them all to determine the method that most successfully reduces imbalance between confounders and exposure.

2a. Create Simplified Weights Formulas Relating Exposure to Confounders at Each Time Point. First, we automatically create shorter, more parsimonious weights formulas relating exposure to confounders at each time point using the `createFormulas()` function, setting *type* = "short." This use of the function automatically creates a shortened formula for each exposure time point. These relate exposure to all user-specified time invariant confounders and only those time-varying confounders at the most proximal prior time point (e.g., for income, only income at 35 months is included for 58-month exposure formula), given that their inclusion can often suffice to balance levels at more distant lags.

Our goal with IPTW weighting is to achieve the greatest reduction in confounder imbalance, with respect to exposure at each time point, using the most parsimonious weights formulas possible. In longitudinal studies with many time points, full weights formulas are long and unwieldy, especially at later time points (e.g., 58 months). Given the relative stability of many constructs over development, balancing only on the confounder at the most recent lag (e.g., income at 35 months) may functionally achieve balance for levels of the confounder at more distal lags (e.g., income at 6, 15, and 24 months). Importantly, we will empirically assess and re-address the validity of this assumption at a subsequent step (Step *3a*). This function outputs shortened weights formulas at each exposure time point (Table 5B) for user inspection. (See Supplement Table s9 for simplified weights formulas for the empirical example).

Figure 6

Example Plot Summarizing Covariate Balance in Relation to Exposure at a Given Time Point



Note. Example summary plot depicting initial confounder imbalance (averaged across imputed datasets) for economic strain exposure at 15 months, prior to weighting. Points represent correlations between confounders and economic strain. Dotted vertical red lines reflect the optional user-specified 0.05 and 0.1 balancing thresholds for critical and less critical confounders, respectively. Balance thresholds are the absolute values below which associations between exposure and confounders are considered negligible. Imbalanced covariates are labeled in red, and in this case, reflect income and maternal depression, resources in the home, and economic strain at 6 months, maternal and child race, government assistance, state, home ownership, maternal marital status, maternal academic skills, and maternal education. Plot output from the `assessBalance()` function. See Supplement Figure s4 for plots at other time points.

2b. Create IPTW Balancing Weights Using Multiple Weighting Methods. Having created shorter, more simplified weights formulas, we now create the first IPTW balancing weights (Cole & Hernán, 2008; Thoemmes & Ong, 2016) using the `createWeights()` function

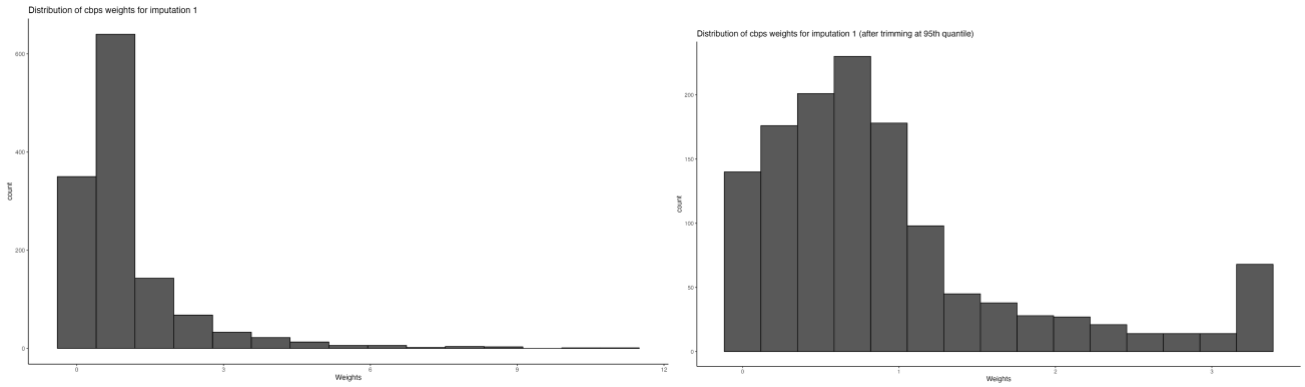
and the shortened weights formulas. This function automatically creates stabilized balancing weights at each time point, and then multiplies the weights across time points (using the *WeightIt* package; Greifer, 2024b), to create a single IPTW weight per person. We recommend users run this function several times, each time supplying a different weighting method in order to create multiple weights and identify the best one. The function outputs summary information and histograms of the weights for user inspection (Figure 7A).

We begin by supplying “cbps" to the *method* field to first create weights using the CBPS method (Huffman & Gameren, 2018). We then iterate through all available weighting methods (see *WeightIt* R package documentation; Greifer, 2024b).

Figure 7

Histograms of IPTW Balancing Weights Prior to and Following Trimming

A. **B.**



Note. (**A)** Example distribution of the original CBPS balancing weights for economic strain exposure (imputation one) with the expected heavy right tail (median = 0.706, sd = 1.227, range = 0 - 11.010). This right skew is unsurprising with real-world data (Box 2), whereby the weights in the far-right tail reflect individuals who experienced statistically unexpected levels of economic strain given their confounder levels: those who were a priori more likely to experience high economic strain but experienced low levels, and/or those who were more likely to experience low economic strain but experienced high levels. Print output from the `createWeights()` function. (**B)** Example distribution of CBPS balancing weights (imputation one) after trimming the original weights or replacing those greater than the 95th quantile with the 95th quantile value. The right heavy tail previously evident in the weights distribution in (A) is now considerably reduced (median = 0.710, SD = 0.090, range = 0 - 3.275). Output from the `trimWeights()` function.

2c. Assess All Weighting Methods to Determine Optimal Method. Next, using the `assessBalance()` function, we evaluate how well each of the weighting methods reduces imbalance in the data. This function automatically provides weighted balance statistics and plots, quantifying the extent to which confounding relations have been successfully minimized in the weighted world (see Supplement and Jackson, 2016). We recommend users run this function multiple times, each time supplying a set of weights estimated with a different method (from the previous step), always retaining the same balance threshold specifications. Next, we recommend that the user manually assess the different outputs of this function to compare the balance of all fitted weights. After inspecting the balance output, we recommend identifying the optimal weighting method as the one that optimally achieves balance by satisfying the following criteria (averaging across imputed datasets, where applicable) of:

- *Fewest imbalanced confounders remaining,* relative to the user-specified balance threshold(s).
- *Lowest median absolute value balance* statistic, across all confounders and time points, reflecting the best overall attenuation of confounding.
- *Lowest maximum absolute balance* statistic, across all confounders and time points (and imputed datasets, where applicable), indicating weakest remaining relation between exposure and confounder for the least balanced confounder.
- *Reasonable effective sample size* following weighting (for all imputed datasets, where applicable), indicating reasonable power to detect effects.

In our empirical example, the CBPS weighting method (Figure 7a; Supplement Figure s5) produced the best balance, successfully balancing 260 of the 261 confounders across all time points and leaving 1 (0.04%) confounder imbalanced. Only income at 6 months remained

imbalanced at 58 months, with a remaining correlation with exposure of -0.0552 (greater than our more stringent absolute value threshold of 0.05) as shown in Supplement Figure s6. The median absolute value correlation between exposure at all time points and confounding variables is 0.06 (max = 0.06). For CBPS, the ESS (for imputation 1) is 540.79 (down from 1,292 in the unweighted sample), indicating that, as is to be expected, we lose some precision with weighting. Of note, the second best-performing method was GBM, with 81 (31%) of covariates remaining imbalanced with a median relation to exposure of 0.15 (max = 0.68), and an ESS of 16.3 (for imputation 1).

Step 3. Create Updated Formulas and Re-Specify Weights using Optimal Weighting Method

The overarching goal of the next step is to create a final round of weights using the optimal weighting method and the simplest yet most effective weights formulas. To do this, we first (*3a*) inspect the balance results from the optimal weighting method (identified in Step 2) and (*3b*) add to the shortened weights formulas any time-varying confounders at lags greater than 1 that were not successfully balanced. We will then use these updated weights formulas to (*3c*) create a final round of weights before (*3d*) trimming, or winsorizing, the weights to reduce any very large values

3a. Examine Balance of Optimal Weighting Method. We recommend users manually closely inspect the balance output of the optimally performing weights from the prior step to get a sense for which confounders remain imbalanced.

3b. Update Simplified Formulas. We then provide the balance statistics we just created to the `createFormulas()` function, setting *type* = “update," to automatically add to the shortened formulas any confounders at longer lags that remain imbalanced, to create a final set of weights formulas. The function outputs updated formulas at each exposure time point (see Table 4) for user inspection.

For example, in our empirical study, we found that income at 6 months was not fully balanced at the last time point (58 months), so this confounder was added to the simplified formula for that time point (see Supplement).

3c. Create Final Balancing Weights. Next, we use the `createWeights()` function to automatically create a final set of balancing weights, by supplying the optimal weighting method identified in Step *2c* and the final, updated formulas from Step *3a*. This function provides summary statistics and histograms of weights for user inspection.

In our example, we used the CBPS weighting method to create a final set of weights using the updated formulas (median = 0.71, sd = 1.21, range = 0 - 10 for imputed dataset 1).

3d. Trim Final Balancing Weights. Subsequently, we consider using the `trimWeights()` function to trim (or winsorize) this final set of weights in order to eliminate the heavy right tail of its distribution (see Box 2). High levels of weight variation, including heavy right tails that are ubiquitous with real-world data (Figure 8a), can lead to wide confidence intervals in subsequent models (Cole & Hernán, 2008). This function replaces the value of all weights greater than the 95th quantile value with the weight value at that quantile (Cole & Hernán, 2008; Kish, 1992) and plots and summarizes the trimmed weights for user inspection.

For our empirical example, we considered trimmed versions of our final weights by trimming them at the 95th quantile, as shown in Figure 7b (median = 0.71, SD = 0.84, range = 0 - 3 for imputed dataset 1).

Step 4. Conduct Final Balance Assessment

Having created and trimmed the final set of balancing weights, the next step is to conduct a final evaluation of how these weights reduce imbalance and the resulting ESS for both trimmed and untrimmed weights. We do this using the `assessBalance()` function, retaining our original balance threshold specifications. This function produces full and summary tables as well as plots for user inspection.

In our example, trimming the weights at the 95th percentile results in one confounder remaining imbalanced: income at 6 months in relation to economic strain at 15 months which remains correlated at -0.0537 (with an overall median remaining absolute correlation absolute value between exposure and confounder of 0.05), and ESS of 712.82. Without trimming, we find no confounders that remain imbalanced but an ESS of 540.79. We choose to retain the trimmed weights as our final weights given that they yield a higher ESS, providing increased precision, and only marginally worse balance (with only one imbalanced confounder that is very close to the balance threshold).

We then recommend that the user carefully manually inspect the results of this balance assessment. With real-world data, it is often the case that some confounders remain imbalanced even after amending the weights formulas and weighting using the best-performing method. Any resulting imbalanced confounders will inform future steps in this process, including substantive interpretation and limitations of findings. We recommend the user manually identify any remaining imbalanced confounders (at any exposure time point) that are time-invariant. The user will have the option to include these confounders as control covariates in the subsequent outcome modeling stage (Step 5). Imbalanced time-varying confounders should *not* be included as control covariates in the modeling stage, as they can actually introduce more bias than they reduce. See Supplement for recommended sensitivity analyses.

In our example, we find that income at 6 months is imbalanced but time-varying and thus cannot be further accounted for through balancing weights, it will remain a potential confounder of our causal association of interest and should be discussed as such.

Phase 2: Assess Substantive Associations between Exposure and Outcome

The goal of this second phase of the MSM computation is to use the IPTW weights from Phase 1, which minimized the associations between exposure and confounders, to assess

substantive hypotheses of the relative roles of dose and timing of exposure across development on the outcome. The steps in this phase constitute the MSM.

Step 5. Fit Weighted Outcome Model & Summarize & Visualize Results

The aim of this final step is to (*5a*) fit a weighted model of the user's choosing, relating exposure at meaningful developmental timepoints to the outcome, before (*5b*) estimating and comparing outcomes across exposure histories, and summarizing and visualizing the results. In turn, the fitted regression parameters for the exposures are used to test specific differences in the model-implied outcomes between specific exposure histories.

5a. Select and Fit a Weighted Outcome Model. Having created and trimmed final IPTW balancing weights that minimize relations between exposure and confounders, we can now select and fit a weighted outcome model. We supply the `fitModel()` function with our final weights, outcome variable, as well as a user-selected, weighted model relating exposure at the user-specified exposure main effects to the outcome. If the user specified exposure epochs in the MSM object, the function will automatically take the average of the exposures specified within each to use the epochs as exposure main effects. If the user did not specify epochs, the function will use exposure at each time point as the exposure main effects. This function fits a weighted generalized linear model (with optional user-specified family and link function) using the supplied weights. The user is required to choose one of the following weighted outcome models:

- *M0:* Baseline model regressing the outcome on the main effects of exposure at each exposure time point or epoch,
- *M1:* Covariate model regressing the outcome on the main effects of exposure, as well as any user-identified covariates (e.g., confounders that are time invariant that remained imbalanced after weighting from Step 4),

- *M2*[12]*:* Interaction model regressing the outcome on the main effects of exposure, as well as interactions between exposure main effects (e.g., “infancy:toddlerhood”) up to a user-specified interaction order, or
- *M3*[12]*:* Full model regressing the outcome on the main effects of exposure, all user-specified covariates, as well as interactions (up to a user-specified interaction order) between exposure main effects and between exposure main effects and covariates.

The function first outputs results from a Wald test assessing whether the model including the exposure variables (see model options above) explains more variance than one without. We recommend users use the results of this test to guide decision-making about conducting further comparisons. The function then outputs a table of model results (which we do not recommend the user interpret).

Given our outcome in the empirical example, which we specify as behavior problems at 58 months or “StrDif_Tot.58” to the *outcome* field, we use a linear outcome model. As discussed earlier, we already specified infancy (“ESETA1.6,” “ESETA1.15”), toddlerhood (“ESETA1.24,” “ESETA1.35”) and childhood (“ESETA1.58”) exposure epochs in the MSM object so these constitute the exposure main effects in the model. We selected model *M0*, as shown below (see Supplement Table s10), with Wald test results of p = 0.092. See Supplement Tables s11-s12 for recommended sensitivity analyses using different quantile values for trimming the weights.

$$StrDif_Tot.58 \sim ESETA1_{infancy} + ESEATA1_{toddlerhood} + ESETA1_{childhood}$$

[12] We note that when using an outcome model with interactions, users may also wish to consider interactions in the weights formulas.

5b. Estimate, Compare, and Visualize Model-Predicted Outcome as a Function of Exposure History. Subsequently, we supply the fitted model results to the `compareHistories()` function that will estimate and compare the predicted average marginal effects of each user-specified exposure history on the outcome. In the case of a continuously distributed exposure, the user is required to specify cutoff values demarcating high and low levels of the exposure to the *hi_lo_cut* field. This simply plugs the proper predictor values into the fitted regression model to estimate the predicted values, conditional on these values, for subsequent comparison.

We also recommend the user specify reference and comparison histories to the *reference* and *comparison* fields, respectively. This function allows us to effectively model the counterfactuals for all possible combinations of high and low economic strain exposure based on the fitted outcome model. `compareHistories()` first estimates predicted values for each user-specified history given the weighted model fitted in the previous step. Next, the function conducts comparison contrasts between user-specified exposure histories, before correcting for multiple comparisons using the default Benjamini-Hochberg method. Please see the ExampleWorkflow.rmd and website Workflow vignettes for more detail on other optional user specifications to this function.

This function produces both a table (e.g., Supplement Table s13) and plot of history comparisons for user inspection. It also provides plots of the model-predicted values for the outcome under each exposure history as shown in Figure 8 (see also Supplement Figures s6-s12. In conjunction with the results from the contrasts (Table 5), these plots provide estimates of the average exposure effects as a function of developmental timing and dosage.

For our empirical example, as discussed prior, we specify 0.60 and 0.30 and percentiles for high and low cutoffs, respectively, for our continuously distributed exposure of economic strain (ESETA1). We conduct the history comparisons delineated in Figure 1.4 to answer our questions investigating infancy as a sensitive period and any dosage effects of exposure to high

economic strain over development. The history comparisons are shown in Table 5 and select examples are plotted for illustrative purposes only in Figure 8. Example sensitivity analyses are shown in Supplement Tables s14-s15 and Tables s16-s17.

Table 5

Example Table of Pairwise Comparisons Comparing the Causal Effects of all User-Specified Exposure Histories

	comparison	estimate	d	SE	statistic	df	p.val	CI.low	CI.high	dose	p.corr	p.corr_all
Timing Questions	(l-l-h) - (h-l-l)	0.018	.06	0.018	1.013	105.818	0.313	-0.017	0.054	1 - 1	0.470	0.470
	(l-h-l) - (h-l-l)	0.033	.11	0.026	1.249	23.138	0.224	-0.022	0.087	1 - 1	0.470	0.378
	(l-l-l) - (h-l-l)	0.006	.02	0.014	0.466	57.422	0.643	-0.021	0.034	0 - 1	0.643	0.697
	(h-l-h) - (l-h-h)	-0.033	-.11	0.026	-1.249	23.140	0.224	-0.087	0.022	2 - 2	0.313	0.378
	(h-h-l) - (l-h-h)	-0.018	-.06	0.018	-1.013	105.842	0.313	-0.054	0.017	2 - 2	0.313	0.470
Dose Questions	(h-l-l) - (l-l-l)	-0.006	-.02	0.014	-0.466	57.422	0.643	-0.034	0.021	1 - 0	0.643	0.697
	(l-h-l) - (l-l-l)	0.026	.09	0.015	1.732	21.603	0.097	-0.005	0.058	1 - 0	0.292	0.306
	(l-l-h) - (l-l-l)	0.012	.04	0.009	1.292	107.579	0.199	-0.006	0.030	1 - 0	0.298	0.378
	(h-h-l) - (h-l-l)	0.026	.09	0.015	1.732	21.597	0.097	-0.005	0.058	2 - 1	0.358	0.306
	(l-h-h) - (h-l-l)	0.045	.15	0.028	1.586	31.184	0.123	-0.013	0.102	2 - 1	0.358	0.332
	(h-l-h) - (h-l-l)	0.012	.04	0.009	1.292	107.576	0.199	-0.006	0.030	2 - 1	0.358	0.378
	(h-h-l) - (l-h-l)	-0.006	-.02	0.014	-0.466	57.454	0.643	-0.034	0.021	2 - 1	0.645	0.697
	(l-h-h) - (l-h-l)	0.012	.04	0.009	1.292	107.569	0.199	-0.006	0.030	2 - 1	0.358	0.378
	(h-l-h) - (l-h-l)	-0.021	-.07	0.028	-0.763	21.964	0.453	-0.078	0.036	2 - 1	0.645	0.644
	(h-h-l) - (l-l-h)	0.008	.03	0.018	0.461	228.661	0.645	-0.027	0.044	2 - 1	0.645	0.697
	(l-h-h) - (l-l-h)	0.026	.09	0.015	1.732	21.599	0.097	-0.005	0.058	2 - 1	0.358	0.306
	(h-l-h) - (l-l-h)	-0.006	-.02	0.014	-0.466	57.455	0.643	-0.034	0.021	2 - 1	0.645	0.697

	(h-h-l) - (h-h-h)	-0.012	-.04	0.009	-1.292	107.598	0.199	-0.030	0.006	2 - 3	0.298	0.378
	(l-h-h) - (h-h-h)	0.006	.02	0.014	0.466	57.448	0.643	-0.021	0.034	2 - 3	0.643	0.697
	(h-l-h) - (h-h-h)	-0.026	-.09	0.015	-1.732	21.601	0.097	-0.058	0.005	2 - 3	0.292	0.306
	(h-h-l) - (l-l-l)	0.020	0.069	0.012	1.638	593.310	0.102	-0.004	0.044	2 - 0	0.153	0.306
	(l-h-h) - (l-l-l)	0.038	0.13	0.017	2.273	32.202	0.030	0.004	0.073	2 - 0	0.089	0.268
	(h-l-h) - (l-l-l)	0.005	0.017	0.015	0.369	41.130	0.714	-0.0024	0.035	2 - 0	0.714	0.714
	(h-l-l) - (h-h-h)	-0.038	-.13	0.017	-2.274	32.199	0.030	-0.073	-0.004	1 - 3	0.089	0.268
	(l-h-l) - (h-h-h)	-0.005	-.02	0.015	-0.369	41.134	0.714	-0.035	0.024	1 - 3	0.714	0.714
	(l-l-h) - (h-h-h)	-0.020	-.07	0.012	-1.638	593.401	0.102	-0.044	0.004	1 - 3	0.153	0.306
	(l-l-l) - (h-h-h)	-0.032	-.11	0.012	-2.615	406.000	0.009	-0.056	-0.008	0 - 3	0.009	0.250

Note. The histories reflect different combinations of high ("h") and low ("l) levels of economic strain over each exposure epoch (infancy, toddlerhood, childhood), resulting in different cumulative doses of exposure to high levels of economic strain. The term corresponds to the unstandardized beta values for each of the terms in the final model: main effects of exposure epoch for infancy, toddlerhood, and childhood for each comparison. Effect sizes (*d*) were calculated manually by dividing the estimate by the standard deviation of the outcome (averaged across imputed datasets; 0.29). Dose counts reflect doses of high exposure. The "p.corr" *p*-values reflect the Benjamini-Hochberg adjustment for false discovery considering each comparison set as a given family (package default), while the "p.corr_all" *p*-values reflect manual adjustment considering all comparisons as a single family. Rows are colored by dose/timing comparison. Output adapted from the `compareHistories()` function.

For our empirical example, we tested two hypotheses about the causal relations between economic strain exposure over development and children's behavior problems, after adjusting for 8 time invariant and 14 time-varying confounders. We note that, as is common with real-world data, we were not able to fully achieve balance for all confounders at all exposure time points. For our final set of weights, we found that income at 6 months remained imbalanced with respect to economic strain at 15 months. This highlights the high concordance between these time-varying constructs and the challenges of fully disambiguating their developmental effects.

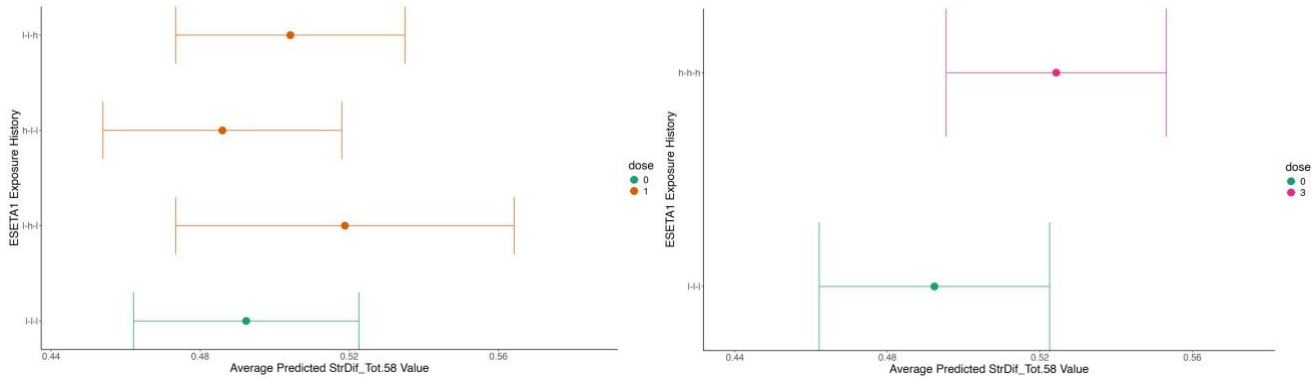
Given that *devMSMs* implements multiple comparison correction separately for each reference-comparison set, we chose to manually correct for multiple comparisons, given that all of our comparisons fall within one family of study questions. We first examined the extent to which the timing of one or two doses of economic strain exposure influenced behavior problems. As shown in Table 5, we found no differences in behavior problems for those who experienced one dose of economic strain during infancy compared to those who experienced one dose in toddlerhood, childhood, or never at all (*p_corr_all*s > 0.30; Supplement Figure s7). Additionally, we found no differences in behavior problems for those who experienced two doses of economic strain across a period including infancy compared to not in infancy (*p_corr_all*s > 0.37; Supplement Figure s8). Second, we examined dose effects of exposure to economic strain in relation to behavior problems and found that no effects survived correction for multiple comparisons (all *p_corr_all*s > 0.25; Supplement Figures s9-s12).

Thus, collectively, we found no support to the hypothesis that experiencing economic strain in infancy was a particularly sensitive developmental span for children's later development of behavior problems. For example, although it failed to reach statistical significance, of those receiving a single dose of economic strain, those who did so in toddlerhood and early childhood showed descriptively modestly worse behavior problems than those experiencing economic strain only in infancy (*d* = .11, *d* = .06, respectively; Figure 8). We also found little support for dosage effects –at least, in linear fashion. For example, prior to adjusting for repeated testing, there was a suggestion that children experiencing chronically high economic strain showed modestly more elevated levels of behavior problems than those with consistently low levels of economic strain (*d* = .13; Figure 8) or those experiencing a single dose in infancy (*d* = .11). There was no consistent evidence indicating that children experiencing two doses of economic strain showed more elevated behavior problems than those who experienced a single dose, nor for a single dose compared to consistently low economic strain. For didactic purposes, we

showcase several different kinds of comparisons to reflect dose and timing questions. However, users could reasonably identify more targeted questions and associated comparisons.

Figure 8

Model-Predicted Behavior Problems as a Function of Exposure Histories Varying in Dose



Note. Weighted outcome MSM model-predicted values of children's behavior problems, comparing *(left)* the different effects histories with 0 (green) vs 1 (orange) dose of exposure; and *(right)* 0 (green) vs 3 (prink) doses of exposure. These comparisons did not survive correction for multiple comparisons when considering all comparisons as a single family (Table 5). As a point of contrast and cautionary note, using an unweighted outcome model that contains biases and does not account for confounding, many of these comparisons survived multiple comparison correction (Supplement Table s19; Figure s13).

As a point of comparison, we provide the same results derived from an unweighted outcome model in Supplement Table s19). Although some of the effect sizes are reasonably similar across the specifications, a noteworthy proportion show substantial differences –the magnitudes of many halving/doubling or more in absolute terms –suggesting that the weighting reduced nontrivial bias due to both the measured confounders and potential effect blocking. It is also worth noting here that there were also instances in which the magnitude of the effects remained similar across specification, yet the precision decreased in the weighted specification.

Indeed, in the absence of weighting, many history comparisons survive multiple comparison corrections. This is somewhat unsurprising given common trade-offs between bias and precision that can occur in the context of IPTW-weighted analyses, where the variance of the weights can impact the variance of the estimates (Chattopadhyay et al., 2020). The ESS of our present sample (averaged across imputations) is 712.82, compared to the total N of 1,292. Thus, the observed differences in the standard errors across the weighted and unweighted specification is largely unsurprising. Importantly, the reduced precision observed for the weighted analyses comes with the benefit of less extrapolation compared to regression adjustment (see Chattopadhyay & Zubizarreta, 2023). It nonetheless raises important future questions concerning the implications of ESS for statistical power in MSMs –a largely unexplored area in the causal inference literature.

8. Discussion & Ongoing Considerations

In summary, this paper provided an overview of how causal concepts can be applied to questions concerning timing and dose using MSMs (Robins et al., 2000). We then introduced how these ideas can be implemented with workflow supported by our open-source R package, *devMSMs*.

8.1 Strengths and Benefits of MSMs via IPTWs

We offer this tutorial on using IPTW for estimating MSMs not as a silver bullet to uncovering causal effects, but rather as an under-utilized tool for improving our causal inference, replete with its own challenges and (often untestable) assumptions. Nonetheless, we believe MSMs via IPTWs offer notable advantages, particularly for developmental scientists working with observational data who are interested in causal questions about dose and timing. MSMs provide a valuable alternative to typical regression-based approaches which, in the presence of time-varying confounders, can introduce more bias than they remove.

In addition to providing a tool for answering causal questions about the effects of dose and timing in development, this approach invites researchers to think deeply about their causal models, establish *a priori* causal hypotheses, and identify their methodological assumptions and limitations. We believe this approach provides a practical tool for those thinking about a host of causal questions, ranging from topics such as basic plasticity mechanisms to social policy. For developmentalists, it requires us to apply greater specificity to our conceptualizations for the way experiences accumulate within individuals to manifest in developmental change. Although this can be a challenging endeavor, it is a valuable one. When our models are strong and specific, IPTWs for estimating MSMs are a powerful tool for adjudicating between alternative explanations. When our models are less developed, these methods highlight regions of the conceptual space that need refinement and provide an opportunity to begin to fill these specific gaps through more exploratory aims. Both are critical to cumulative developmental science.

Here, we illustrated how MSMs may offer new insights into one specific developmental question: whether, when, and to what extent exposure to household economic strain causes children's subsequent behavior problems. Despite prior work suggesting that economic strain may undermine children's subsequent social development (Burchinal et al., 2008; Conger et al., 1994) as well as work suggesting that this may be particularly the case when experienced in infancy (e.g., Brandes-Aitken et al., 2019; Combs-Orme & Cain, 2006; Frick et al., 2018; Hanson et al., 2013; Ramanathan et al., 2017), we found little evidence of either, after ruling out a host of time-invariant and time-varying confounders. While future replication in larger and more socioeconomically diverse samples is warranted, we found no support for infancy serving as a particularly salient developmental period. Further, we found minimal support for a dosage effect. After adjusting for a battery of confounders, we found some indication that children experiencing three consecutive doses showed modestly more elevated behavior problems, on average, than those who had experienced continuously low economic strain or those who experienced a single dose in infancy. This could suggest that two consecutive doses of

economic strain in toddlerhood and early childhood may lead to modestly more elevated behavior problems. Notably, these effects did not survive corrections for multiple testing. Otherwise, we found little consistent evidence that experiencing two or a single dose of economic strain was predictive of elevated behavior problems relative to those experiencing fewer doses.

These discrepant findings could be explained by several factors. Consistent with the aims of the approach, the discrepancies are quite possibly explained by the fact that, by balancing many time-invariant and time-varying confounders across exposure histories, we removed a number of biasing backdoor pathways in a manner that prior work has not. Indeed, with respect to time-varying confounders, our balancing approach accounts for these backdoor pathways in ways that could not be captured using typical regression adjustment. Of course, the discrepancies could also be explained by other factors, including varying measures of economic strain and behavior problems, as well as differences in the underlying populations. For example, our predominantly lower-income sample may have shown less variability in economic strain than the general population. The discrepancies could also be partially explained by bias-precision tradeoffs that can manifest with IPTW-based approaches, as described earlier. Importantly, a biased estimate estimated with precision is not better than an unbiased but less precise estimate, where the imprecision reflects real uncertainty about the unobserved counterfactual.

We find that one of the advantages of adopting the potential outcome framework is that it disambiguates the conceptual causal effect of interest –including what constitutes an effective dose– from the statistical model used to estimate that causal effect. It forces the researcher to think carefully about both the substantive causal model motivating the work, as well as the way that substantive model connects to their empirical tests ‘on the ground.’ As such, it aligns well with increasing calls for greater clarity and specificity in psychological science (e.g., Bringmann

et al., 2022; Frankenhuis et al., 2023; Meehl, 1967; Muthukrishna & Henrich, 2019) and helps to support a common language for results across a literature.

8.2 Limitations of MSMs

Despite the benefits of MSMs via IPTWs for time-varying exposures, the approach also has limitations. First, the internal validity of the inferences rests on meeting several assumptions, some of which are impossible to establish empirically. These assumptions are critical to attributing causality to estimated associations. Yet, in most real-world scenarios, they are difficult to meet, and we are left to make the strongest case we can, given the data we have.

Causal diagrams (e.g., DAGs) serve as useful tools to support this process, though they bring their own challenges. Creating conceptual distinctions between confounders, colliders, and mediators within a complex developmental causal pathway model with potentially hundreds of measured variables is daunting. The number of decision points in this effort can become unwieldy and somewhat arbitrary, particularly when our theoretical models are underdeveloped and/or the particularities of one's data lead to ambiguities in the causal model. For example, we see this for the contemporaneously measured time-varying exposures and confounders in our DAG 2. We chose not to treat contemporaneously measured maternal depression as a confounder because the temporal structure and theoretical ambiguity precluded distinguishing whether it acted as a confounder or mediator. We erred on the side of avoiding introducing bias, but we cannot know with certainty whether this was the right decision. However, in situations like these, the strength of DAGs is to make these choices explicit, so that their implications can be interrogated. For example, we could conduct sensitivity analyses to consider how our conclusions would change if we erred in the other direction. Thus, we have made our assumptions clear –for ourselves and others.

In the context of continuous exposures, researchers face the additional challenge of determining what 'high' and 'low' values of the exposure functionally serve as an effective single dose. When this cannot be grounded in theory, it can become a more arbitrary blend of

convention and the practicalities of one's data. For example, in our own data, although we originally intended to use the 75th and 25th percentiles as our high and low threshold, respectively, we found that this led to some exposure histories with rather sparse representation in the multivariate space of the time-varying exposure. Because this can affect the precision of the estimated counterfactual differences, we subsequently adjusted the thresholds to capture a better representation of individuals in our data. Although these decisions were made prior to testing the outcome model, they introduce additional researcher degrees of freedom that need to be documented and defended. Notably, these types of decisions are not unique to MSMs – just more explicit. We do this every time we probe the simple slopes of a statistical interaction at 'high' and 'low' values of a continuous moderator. It is often even implicit in the creation of binary exposure variables (e.g., how many doses of an intervention/drug trial constitute an 'effective dose?'). We find the fact that using MSMs forces one to make these assumptions explicit to be a strength.

Lastly, MSMs cannot represent all kinds of causal concepts (Ross, 2021). For instance, as many developmental frameworks assert, some causal relations may arise from more complex, nonlinear, emergent processes involving interaction-dominant feedback loops between interrelated variables (Hasselman, 2022; Smith & Thelen, 2003). This type of causal process is beyond the scope of the typical 'machine-like' or component-dominant framework of causal mechanisms for which MSMs model.

8.3 Future Directions

The rate of innovation since the early emergence of IPTW for MSMs (e.g., Robins, 1986; Robins et al., 2000) has been truly remarkable. Scholars of causal inference continue to provide new ways of thinking about causation in the context of time-varying exposures, as well as increasingly complex computational tools to test and evaluate them (Coulombe & Yang, 2024; Mulder et al., 2024; Zhou & Wodtke, 2020). Our aim here was to present an approachable overview of the method for applied researchers working with time-varying exposures and

entertaining questions concerning timing and dose. As such, we have only scratched the surface. For example, the methods we described can be readily extended to time-varying outcomes (e.g., Blackwell, 2013; Thoemmes & Ong, 2016; Vandecandelaere et al., 2016) and other applications, such as causal mediation (e.g., Coffman & Zhong, 2012; Moerkerke et al., 2015; VanderWeele, 2009). We aspire to add increasing functionality to the *devMSMs* package.

Of course, innovation requires continued refinement. For instance, while IPTW approaches for continuous exposures have been available for some time (e.g., Hirano & Imbens, 2004; Imai & van Dyk, 2004) and ongoing work continues to optimize the method for time-invariant continuous exposures (e.g., Fong et al., 2018; Huling et al., 2023), their application to estimating unbiased MSMs has been somewhat understudied. Similarly, with real-world data, there will often be times when the available data do not perfectly align with our theoretical models for the causal process of interest, leaving applied researchers somewhat stymied about the best way to move forward. In the current example, for instance, our theoretical model was not strong enough to support hypotheses beyond broad developmental spans. To address this, we devised IPTWs based on all the available exposures and then subsequently collapsed them into substantively meaningful epochs in our outcome model. This was a practical decision, based on the intuition that creating balance across all the exposures would prove more effective for removing bias than balancing for the epochs or simply ignoring the existence of some measured exposures. The impact of these practical solutions requires further testing. Another example of this ambiguity can be found in DAG structures, when some of the causal directions remain unclear. Indeed, although more data-driven exploratory work with DAGs is beginning to emerge (e.g., Tennant et al., 2021) it remains somewhat unclear how applied researchers should navigate these ambiguities.

Lastly, as we have alluded above, IPTWs can manifest in trade-offs between bias and precision (e.g., Chattopadhyay et al., 2020, 2023; Hazlett & Shinkre, 2024). This can have implications for statistical power. Future research needs to use the bias-reducing methods we

describe while providing enough data to estimate effects with precision. Power analysis needs to account for the inherent uncertainty of estimating missing potential outcomes without extrapolating. Unfortunately, power analysis for MSMs has only recently begun to emerge (see Shook-Sa & Hudgens, 2022) and can be quite challenging. One of the challenges is that it requires an *a priori* sense of the *design effect* –or the variance in the estimate induced by the variability of the weights between individuals. This is difficult to know. Although Shook-Sa and Hudgens (2022) provide some guidelines for approximating design effects, further development is needed.

9. Conclusion

Taken together, the present tutorial aimed to provide an approachable introduction to IPTW for MSMs for testing questions concerning the causal effects of timing and dose of exposures on later outcomes. We developed a user-friendly R package, *devMSMs,* to support some of the more common workflows for estimating MSMs using IPTWs and to facilitate greater adoption of the method amongst applied psychological researchers. Additional detailed vignettes are provided online for interested users. Although MSMs –like all statistical approaches –come with important limitations and areas in need of refinement, the strengths we have highlighted here make them a useful tool, particularly for developmental scientists. By triangulating across empirical findings using MSMs and other causally informed methods (e.g., Bailey et al., 2024), we can enhance the rigor and credibility of our research. This approach has far-reaching implications for both science and policy: it allows us to build a more comprehensive understanding of how and when exposures impact development and enables more strategic resource allocation to programs targeting critical developmental windows. We hope the broader adoption of MSMs helps to bridge the gap between rigorous scientific inquiry and actionable implementation, driving meaningful change in real-world settings.

References

Arel-Bundock, V., Diniz, M. A., Greifer, N., & Bacher, E. (2024). *marginaleffects: Predictions, Comparisons, Slopes, Marginal Means, and Hypothesis Tests* (Version 0.12.0) [Computer software]. https://cran.r-project.org/web/packages/marginaleffects/index.html

Austin, P. C. (2011). An Introduction to Propensity Score Methods for Reducing the Effects of Confounding in Observational Studies. *Multivariate Behavioral Research*, *46*(3), 399–424. https://doi.org/10.1080/00273171.2011.568786

Austin, P. C. (2019). Assessing covariate balance when using the generalized propensity score with quantitative or continuous exposures. *Statistical Methods in Medical Research*, *28*(5), 1365–1377. https://doi.org/10.1177/0962280218756159

Bagner, D. M., Pettit, J. W., Lewinsohn, P. M., & Seeley, J. R. (2010). Effect of Maternal Depression on Child Behavior: A Sensitive Period? *Journal of the American Academy of Child & Adolescent Psychiatry*, *49*(7), 699–707. https://doi.org/10.1016/j.jaac.2010.03.012

Bailey, D. H., Jung, A. J., Beltz, A. M., Eronen, M. I., Gische, C., Hamaker, E. L., Kording, K. P., Lebel, C., Lindquist, M. A., Moeller, J., Razi, A., Rohrer, J. M., Zhang, B., & Murayama, K. (2024). Causal inference on human behaviour. *Nature Human Behaviour*, *8*(8), 1448–1459. https://doi.org/10.1038/s41562-024-01939-z

Beck, C. T. (1999). Maternal depression and child behaviour problems: A meta-analysis. *Journal of Advanced Nursing*, *29*(3), 623–629. https://doi.org/10.1046/j.1365-2648.1999.00943.x

Blackwell, M. (2013). A Framework for Dynamic Causal Inference in Political Science. *American Journal of Political Science*, *57*(2), 504–520. https://doi.org/10.1111/j.1540-5907.2012.00626.x

Blair, C., Granger, D. A., Willoughby, M., Mills-Koonce, R., Cox, M., Greenberg, M. T., Kivlighan, K. T., & Fortunato, C. K. (2011). Salivary Cortisol Mediates Effects of Poverty

and Parenting on Executive Functions in Early Childhood. *Child Development*, *82*(6), 1970–1984. https://doi.org/10.1111/j.1467-8624.2011.01643.x

Blair, C., & Raver, C. C. (2016). Poverty, Stress, and Brain Development: New Directions for Prevention and Intervention. *Academic Pediatrics*, *16*(3), S30–S36. https://doi.org/10.1016/j.acap.2016.01.010

Boos, D. D., & Stefanski, L. A. (2013). *Essential Statistical Inference* (Vol. 120). Springer New York. https://doi.org/10.1007/978-1-4614-4818-1

Brandes-Aitken, A., Braren, S., Swingler, M., Voegtline, K., & Blair, C. (2019). Sustained attention in infancy: A foundation for the development of multiple aspects of self-regulation for children in poverty. *Journal of Experimental Child Psychology*, *184*, 192–209. https://doi.org/10.1016/j.jecp.2019.04.006

Bray, B. C., Almirall, D., Zimmerman, R. S., Lynam, D., & Murphy, S. A. (2006). Assessing the total effect of time-varying predictors in prevention research. *Prevention Science: The Official Journal of the Society for Prevention Research*, *7*(1), 1–17. https://doi.org/10.1007/s11121-005-0023-0

Bringmann, L. F., Elmer, T., & Eronen, M. I. (2022). Back to Basics: The Importance of Conceptual Clarification in Psychological Science. *Current Directions in Psychological Science*, *31*(4), 340–346. https://doi.org/10.1177/09637214221096485

Bronfenbrenner, U., & Ceci, S. J. (1994). Nature-nuture reconceptualized in developmental perspective: A bioecological model. *Psychological Review*, *101*, 568–586. https://doi.org/10.1037/0033-295X.101.4.568

Burchinal, M., Howes, C., Pianta, R., Bryant, D., Early, D., Clifford, R., & Barbarin, O. (2008). Predicting Child Outcomes at the End of Kindergarten from the Quality of Pre-Kindergarten Teacher–Child Interactions and Instruction. *Applied Developmental Science*, *12*(3), 140–153. https://doi.org/10.1080/10888690802199418

Buuren, S. van, & Groothuis-Oudshoorn, K. (2011). mice: Multivariate Imputation by Chained Equations in R. *Journal of Statistical Software*, *45*(1), 1–67. https://doi.org/10.18637/jss.v045.i03

Chattopadhyay, A., Cohn, E. R., & Zubizarreta, J. R. (2023). *One-Step weighting to generalize and transport treatment effect estimates to a target population* (arXiv:2203.08701). arXiv. https://doi.org/10.48550/arXiv.2203.08701

Chattopadhyay, A., Hase, C. H., & Zubizarreta, J. R. (2020). Balancing vs modeling approaches to weighting in practice. *Statistics in Medicine*, *39*(24), 3227–3254. https://doi.org/10.1002/sim.8659

Chattopadhyay, A., & Zubizarreta, J. R. (2023). On the implied weights of linear regression for causal inference. *Biometrika*, *110*(3), 615–629. https://doi.org/10.1093/biomet/asac058

Coffman, D. L., & Zhong, W. (2012). Assessing mediation using marginal structural models in the presence of confounding and moderation. *Psychological Methods*, *17*(4), 642–664. https://doi.org/10.1037/a0029311

Cole, S. R., & Hernán, M. A. (2008). Constructing Inverse Probability Weights for Marginal Structural Models. *American Journal of Epidemiology*, *168*(6), 656–664. https://doi.org/10.1093/aje/kwn164

Cole, S. R., Platt, R. W., Schisterman, E. F., Chu, H., Westreich, D., Richardson, D., & Poole, C. (2010). Illustrating bias due to conditioning on a collider. *International Journal of Epidemiology*, *39*(2), 417–420. https://doi.org/10.1093/ije/dyp334

Combs-Orme, T., & Cain, D. S. (2006). Poverty and the Daily Lives of Infants. *Journal of Children and Poverty*, *12*(1), 1–20. https://doi.org/10.1080/10796120500502052

Conger, R. D., Ge, X., Elder, G. H., Lorenz, F. O., & Simons, R. L. (1994). Economic Stress, Coercive Family Process, and Developmental Problems of Adolescents. *Child Development*, *65*(2), 541–561. https://doi.org/10.1111/j.1467-8624.1994.tb00768.x

Coulombe, J., & Yang, S. (2024). Multiply robust estimation of marginal structural models in observational studies subject to covariate-driven observations. *Biometrics*, *80*(3), ujae065. https://doi.org/10.1093/biomtc/ujae065

Daniel, R. m., Cousens, S. n., De Stavola, B. l., Kenward, M. G., & Sterne, J. a. C. (2013). Methods for dealing with time-dependent confounding. *Statistics in Medicine*, *32*(9), 1584–1618. https://doi.org/10.1002/sim.5686

de Quidt, J., & Haushofer, J. (2016). *Depression for Economists* (NBER Working Paper 22973). National Bureau of Economic Research, Inc. https://econpapers.repec.org/paper/nbrnberwo/22973.htm

DeJoseph, M. L., Sifre, R. D., Raver, C. C., Blair, C. B., & Berry, D. (2021). Capturing Environmental Dimensions of Adversity and Resources in the Context of Poverty Across Infancy Through Early Adolescence: A Moderated Nonlinear Factor Model. *Child Development*, *n/a*(n/a). https://doi.org/10.1111/cdev.13504

Ettman, C. K., Fan, A. Y., Philips, A. P., Adam, G. P., Ringlein, G., Clark, M. A., Wilson, I. B., Vivier, P. M., & Galea, S. (2023). Financial strain and depression in the U.S.: A scoping review. *Translational Psychiatry*, *13*(1), 1–17. https://doi.org/10.1038/s41398-023-02460-z

Fong, C., Hazlett, C., & Imai, K. (2018). Covariate balancing propensity score for a continuous treatment: Application to the efficacy of political advertisements. *The Annals of Applied Statistics*, *12*(1), 156–177. https://doi.org/10.1214/17-AOAS1101

Fong, C., Ratkovic, M., Imai, K., Hazlett, C., Yang, X., & Peng, S. (2019). *CBPS: Covariate Balancing Propensity Score* (Version 0.21) [Computer software]. https://CRAN.R-project.org/package=CBPS

Foster, E. M. (2010). Causal inference and developmental psychology. *Developmental Psychology*, *46*(6), 1454–1480. https://doi.org/10.1037/a0020204

Frankenhuis, W. E., Panchanathan, K., & Smaldino, P. E. (2023). Strategic Ambiguity in the Social Sciences. *Social Psychological Bulletin*, *18*, 1–25. https://doi.org/10.32872/spb.9923

Frick, M. A., Forslund, T., Fransson, M., Johansson, M., Bohlin, G., & Brocki, K. C. (2018). The role of sustained attention, maternal sensitivity, and infant temperament in the development of early self-regulation. *British Journal of Psychology*, *109*(2), 277–298. https://doi.org/10.1111/bjop.12266

Goodman, R. (2001). Psychometric Properties of the Strengths and Difficulties Questionnaire. *Journal of the American Academy of Child & Adolescent Psychiatry*, *40*(11), 1337–1345. https://doi.org/10.1097/00004583-200111000-00015

Greenland, S., Pearl, J., & Robins, J. M. (1999). Causal Diagrams for Epidemiologic Research. *Epidemiology*, *10*(1), 37–48.

Greifer, N. (2024a). *cobalt: Covariate Balance Tables and Plots* (Version 4.4.1) [Computer software]. https://CRAN.R-project.org/package=cobalt

Greifer, N. (2024b). *WeightIt: Weighting for Covariate Balance in Observational Studies* (Version 0.13.1) [Computer software]. https://CRAN.R-project.org/package=WeightIt

Greifer, N., & Stuart, E. A. (2021, June 19). *Choosing the Causal Estimand for Propensity Score Analysis of Observational Studies*. arXiv.Org. https://arxiv.org/abs/2106.10577v2

Haber, N. A., Wood, M. E., Wieten, S., & Breskin, A. (2022). DAG With Omitted Objects Displayed (DAGWOOD): A framework for revealing causal assumptions in DAGs. *Annals of Epidemiology*, *68*, 64–71. https://doi.org/10.1016/j.annepidem.2022.01.001

Haft, S. L., & Hoeft, F. (2017). Poverty's Impact on Children's Executive Functions: Global Considerations. *New Directions for Child and Adolescent Development*, *2017*(158), 69–79. https://doi.org/10.1002/cad.20220

Hanson, J. L., Hair, N., Shen, D. G., Shi, F., Gilmore, J. H., Wolfe, B. L., & Pollak, S. D. (2013). Family Poverty Affects the Rate of Human Infant Brain Growth. *PLOS ONE*, *8*(12), e80954. https://doi.org/10.1371/journal.pone.0080954

Hasselman, F. (2022). *The Complex Systems Approach to Behavioural Science*. https://complexity-methods.github.io/book/

Hazlett, C., & Shinkre, T. (2024). *Understanding and avoiding the "weights of regression": Heterogeneous effects, misspecification, and longstanding solutions* (arXiv:2403.03299). arXiv. https://doi.org/10.48550/arXiv.2403.03299

Hernán, M., Brumback, B., & Robins, J. (2000). Marginal Structural Models to Estimate the Causal Effect of Zidovudine on the Survival of HIV-Positive Men. *Epidemiology*, *11*(5), 561.

Hernán, M., Hernández-Díaz, S., & Robins, J. (2004). A Structural Approach to Selection Bias. *Epidemiology*, *15*(5), 615–625.

Hernán, M., Hernández-Díaz, S., Werler, M., & Mitchell, A. (2002). Causal Knowledge as a Prerequisite for Confounding Evaluation: An Application to Birth Defects Epidemiology. *American Journal of Epidemiology*, *155*(2), 176–184. https://doi.org/10.1093/aje/155.2.176

Hernán, M., & Robins, J. (2020). *Causal Inference: What If*. Chapman & Hall/CRC.

Hernán, M., & Robins, J. (2024). *Causal Inference: What If*. CRC Press.

Hirano, K., & Imbens, G. W. (2004). The Propensity Score with Continuous Treatments. In *Applied Bayesian Modeling and Causal Inference from Incomplete-Data Perspectives* (pp. 73–84). John Wiley & Sons, Ltd. https://doi.org/10.1002/0470090456.ch7

Huffman, C., & Gameren, E. van. (2018). Covariate Balancing Inverse Probability Weights for Time-Varying Continuous Interventions. *Journal of Causal Inference*, *6*(2). https://doi.org/10.1515/jci-2017-0002

Huling, J. D., Greifer, N., & Chen, G. (2023). Independence weights for causal inference with continuous treatments. *Journal of the American Statistical Association*, *0*(ja), 1–25. https://doi.org/10.1080/01621459.2023.2213485

Hurwich-Reiss, E., Watamura, S. E., Raver, C. C., Berlin, L., Blair, C., Constantino, J. N., Hallam, R. A., Han, M., Hustedt, J. T., Harden, B. J., Sarche, M., Vu, J. A., & The BTS Consortium PI's. (2019). Beyond Income: Expanding our Empirical Toolkit to Better Predict Caregiver Well-Being. *Journal of Child and Family Studies*, *28*(3), 753–764. https://doi.org/10.1007/s10826-018-01304-5

Imai, K., & Ratkovic, M. (2014). Covariate balancing propensity score. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, *76*(1), 243–263. https://doi.org/10.1111/rssb.12027

Imai, K., & van Dyk, D. A. (2004). Causal Inference With General Treatment Regimes. *Journal of the American Statistical Association*, *99*(467), 854–866. https://doi.org/10.1198/016214504000001187

Imbens, G. W., & Rubin, D. B. (2015). *Causal Inference in Statistics, Social, and Biomedical Sciences*. Cambridge University Press.

Jackson, J. W. (2016). Diagnostics for Confounding of Time-varying and Other Joint Exposures. *Epidemiology*, *27*(6), 859. https://doi.org/10.1097/EDE.0000000000000547

Kendler, K. S., Karkowski, L. M., & Prescott, C. A. (1999). Causal relationship between stressful life events and the onset of major depression. *The American Journal of Psychiatry*, *156*(6), 837–841. https://doi.org/10.1176/ajp.156.6.837

Kerker, B. D., Zhang, J., Nadeem, E., Stein, R. E. K., Hurlburt, M. S., Heneghan, A., Landsverk, J., & McCue Horwitz, S. (2015). Adverse Childhood Experiences and Mental Health, Chronic Medical Conditions, and Development in Young Children. *Academic Pediatrics*, *15*(5), 510–517. https://doi.org/10.1016/j.acap.2015.05.005

Kish, L. (1992). Weighting for Unequal Pi. *Journal of Official Statistics*, *8*(2), 183.

Knudsen, E. I. (2004). Sensitive Periods in the Development of the Brain and Behavior. *Journal of Cognitive Neuroscience*, *16*(8), 1412–1425. https://doi.org/10.1162/0898929042304796

Leyrat, C., Carpenter, J. R., Bailly, S., & Williamson, E. J. (2021). Common Methods for Handling Missing Data in Marginal Structural Models: What Works and Why. *American Journal of Epidemiology*, *190*(4), 663–672. https://doi.org/10.1093/aje/kwaa225

Loh, W. W. (2024). Estimating curvilinear time-varying treatment effects: Combining g-estimation of structural nested mean models with time-varying effect models for longitudinal causal inference. *Psychological Methods*. https://doi.org/10.1037/met0000637

Loh, W. W., Ren, D., & West, S. G. (2024). Parametric g-formula for Testing Time-Varying Causal Effects: What It Is, Why It Matters, and How to Implement It in Lavaan. *Multivariate Behavioral Research*, *59*(5), 995–1018. https://doi.org/10.1080/00273171.2024.2354228

McEWEN, B. S., & Seeman, T. (1999). Protective and Damaging Effects of Mediators of Stress: Elaborating and Testing the Concepts of Allostasis and Allostatic Load. *Annals of the New York Academy of Sciences*, *896*(1), 30–47. https://doi.org/10.1111/j.1749-6632.1999.tb08103.x

Meehl, P. E. (1967). Theory-Testing in Psychology and Physics: A Methodological Paradox. *Philosophy of Science*, *34*(2), 103–115. https://doi.org/10.1086/288135

Moerkerke, B., Loeys, T., & Vansteelandt, S. (2015). Structural equation modeling versus marginal structural modeling for assessing mediation in the presence of posttreatment confounding. *Psychological Methods*, *20*(2), 204–220. https://doi.org/10.1037/a0036368

Mulder, J., Luijken, K., Vries, B., & Hamaker, E. (2024). Causal Effects of Time-Varying Exposures: A Comparison of Structural Equation Modeling and Marginal Structural

Models in Cross-Lagged Panel Research. *Structural Equation Modeling: A Multidisciplinary Journal*. https://doi.org/10.1080/10705511.2024.2316586

Muthukrishna, M., & Henrich, J. (2019). A problem in theory. *Nature Human Behaviour*, *3*(3), Article 3. https://doi.org/10.1038/s41562-018-0522-1

Naimi, A. I., Moodie, E. E. M., Auger, N., & Kaufman, J. S. (2014). Semiparametric Adjusted Exposure-Response Curves. *Epidemiology*, *25*(6), 919. https://doi.org/10.1097/EDE.0000000000000158

Pearl, J. (1995). Causal diagrams for empirical research. *Biometrika*, *82*(4), 669–688. https://doi.org/10.1093/biomet/82.4.669

Price, R. H., Choi, J. N., & Vinokur, A. D. (2002). Links in the chain of adversity following job loss: How financial strain and loss of personal control lead to depression, impaired functioning, and poor health. *Journal of Occupational Health Psychology*, *7*(4), 302–312. https://doi.org/10.1037/1076-8998.7.4.302

Ramanathan, S., Balasubramanian, N., & Faraone, S. V. (2017). Familial transient financial difficulties during infancy and long-term developmental concerns. *Psychological Medicine*, *47*(12), 2197–2204. https://doi.org/10.1017/S0033291717000666

Raver, C. C. (2012). Low-Income Children's Self-Regulation in the Classroom: Scientific Inquiry for Social Change. *The American Psychologist*, *67*(8), 681–689. https://doi.org/10.1037/a0030085

Robins, J. (1986). A new approach to causal inference in mortality studies with a sustained exposure period—Application to control of the healthy worker survivor effect. *Mathematical Modelling*, *7*(9), 1393–1512. https://doi.org/10.1016/0270-0255(86)90088-6

Robins, J., & Hernán, M. (2008). Estimation of the causal effects of time-varying exposure. In *Longitudinal Data Analysis* (pp. 553–599). https://doi.org/10.1201/9781420011579.ch23

Robins, J., Hernán, M., & Brumback, B. (2000). Marginal Structural Models and Causal Inference in Epidemiology. *Epidemiology*, *11*(5), 550–560.

Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, *70*(1), 41–55. https://doi.org/10.1093/biomet/70.1.41

Ross, L. N. (2021). Causal Concepts in Biology: How Pathways Differ from Mechanisms and Why It Matters. *The British Journal for the Philosophy of Science*, *72*(1), 131–158. https://doi.org/10.1093/bjps/axy078

Ross, R., Zivich, P., Stringer, J., & Cole, S. (2024). M-estimation for common epidemiological measures: Introduction and applied examples. *International Journal of Epidemiology*, *53*(2), dyae030. https://doi.org/10.1093/ije/dyae030

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, *66*(5), 688–701. https://doi.org/10.1037/h0037350

Rubin, D. B. (1976). Inference and missing data. *Biometrika*, *63*(3), 581–592. https://doi.org/10.1093/biomet/63.3.581

Rubin, D. B. (1978). Bayesian Inference for Causal Effects: The Role of Randomization. *The Annals of Statistics*, *6*(1), 34–58.

Rubin, D. B. (2001). Using Propensity Scores to Help Design Observational Studies: Application to the Tobacco Litigation. *Health Services and Outcomes Research Methodology*, *2*(3), 169–188. https://doi.org/10.1023/A:1020363010465

Rubin, D. B. (2005). Causal Inference Using Potential Outcomes: Design, Modeling, Decisions. *Journal of the American Statistical Association*, *100*(469), 322–331. https://doi.org/10.1198/016214504000001880

Shook-Sa, B. E., & Hudgens, M. G. (2022). Power and Sample Size for Observational Studies of Point Exposure Effects. *Biometrics*, *78*(1), 388–398. https://doi.org/10.1111/biom.13405

Smith, L. B., & Thelen, E. (2003). Development as a dynamic system. *Trends in Cognitive Sciences*, *7*(8), 343–348. https://doi.org/10.1016/S1364-6613(03)00156-6

Stolove, C. A., Galatzer-Levy, I. R., & Bonanno, G. A. (2017). Emergence of depression following job loss prospectively predicts lower rates of reemployment. *Psychiatry Research*, *253*, 79–83. https://doi.org/10.1016/j.psychres.2017.03.036

Stuart, E. A. (2008). Developing practical recommendations for the use of propensity scores: Discussion of 'A critical appraisal of propensity score matching in the medical literature between 1996 and 2003' by Peter Austin, Statistics in Medicine. *Statistics in Medicine*, *27*(12), 2062–2065. https://doi.org/10.1002/sim.3207

Tennant, P. W. G., Murray, E. J., Arnold, K. F., Berrie, L., Fox, M. P., Gadd, S. C., Harrison, W. J., Keeble, C., Ranker, L. R., Textor, J., Tomova, G. D., Gilthorpe, M. S., & Ellison, G. T. H. (2021). Use of directed acyclic graphs (DAGs) to identify confounders in applied health research: Review and recommendations. *International Journal of Epidemiology*, *50*(2), 620–632. https://doi.org/10.1093/ije/dyaa213

Textor, J. (2015). *Drawing and Analyzing Causal DAGs with DAGitty*.

Thoemmes, F., & Ong, A. D. (2016). A Primer on Inverse Probability of Treatment Weighting and Marginal Structural Models. *Emerging Adulthood*, *4*(1), 40–59. https://doi.org/10.1177/2167696815621645

Valletta, R. (2015). Higher Education, Wages, and Polarization. . . *Top*.

Vandecandelaere, M., Vansteelandt, S., De Fraine, B., & Van Damme, J. (2016). Time-Varying Treatments in Observational Studies: Marginal Structural Models of the Effects of Early Grade Retention on Math Achievement. *Multivariate Behavioral Research*, *51*(6), 843–864. https://doi.org/10.1080/00273171.2016.1155146

VanderWeele, T. J. (2009). Marginal structural models for the estimation of direct and indirect effects. *Epidemiology (Cambridge, Mass.)*, *20*(1), 18–26. https://doi.org/10.1097/EDE.0b013e31818f69ce

Vernon-Feagans, L., Cox, M., Willoughby, M., Burchinal, M., Garrett-Peters, P., Mills-Koonce, R., Garrett-Peiers, P., Conger, R. D., & Bauer, P. J. (2013). THE FAMILY LIFE PROJECT: AN EPIDEMIOLOGICAL AND DEVELOPMENTAL STUDY OF YOUNG CHILDREN LIVING IN POOR RURAL COMMUNITIES. *Monographs of the Society for Research in Child Development*, *78*(5), i–150.

Woodward, J. (2005). *Making Things Happen: A Theory of Causal Explanation*. Oxford University Press, USA.

Yarkoni, T. (2022). The generalizability crisis. *Behavioral and Brain Sciences*, *45*, e1. https://doi.org/10.1017/S0140525X20001685

Zhou, X., & Wodtke, G. T. (2020). Residual Balancing: A Method of Constructing Weights for Marginal Structural Models. *Political Analysis*, *28*(4), 487–506. https://doi.org/10.1017/pan.2020.2

Zivich, P. N., Cole, S. R., & Westreich, D. (2022). *Positivity: Identifiability and Estimability* (arXiv:2207.05010). arXiv. https://doi.org/10.48550/arXiv.2207.05010